Electromagnetic Waves

Question1

An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10\,\mathrm{mho/m}$. The ratio of maximum conduction current density to maximum displacement current density is ____.

$$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\,\mathrm{Nm^2/C^2}\right]$$

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Answer: 1800

Solution:

According to Ohm's Law in point form, the conduction current density is proportional to the electric field (E) :

$\mathrm{J_c} = \sigma\mathrm{E}$

The maximum value occurs when the electric field is at its peak $(\mathrm{E_0})$:

$(\mathrm{J_c})_{\max} = \sigma\mathrm{E_0}$

Displacement current density arises from a time-varying electric field. For an electromagnetic wave, the electric field can be represented as $E = E_0 \sin(\omega t)$.

Maxwell defined displacement current density as :

$\mathrm{J_d} = \epsilon\frac{\partial\mathrm{E}}{\partial\mathrm{t}}$

$\Rightarrow \mathrm{J_d} = \epsilon\frac{\partial}{\partial\mathrm{t}}(\mathrm{E_0}\sin(\omega\mathrm{t})) = \epsilon\omega\mathrm{E_0}\cos(\omega\mathrm{t})$

The maximum value occurs when the cosine term is 1 :

$(\mathrm{J_d})_{\max} = \epsilon\omega\mathrm{E_0}$

The ratio of the maximum values is:

Ratio $= \frac{(J_c)_{max}}{(J_d)_{max}} = \frac{\sigma E_0}{\epsilon\omega E_0} = \frac{\sigma}{\epsilon\omega}$

The conductivity (σ) of the medium is, $\sigma = 10\text{mho/m}$

The angular frequency (ω) is $2\pi f = 2\pi \times \left(100 \times 10^6\right) = 2\pi \times 10^8 \text{rad/s}$

Since no specific medium permittivity is given, we use the permittivity of free space (ϵ_0).

Given $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$, then $\epsilon_0 = \frac{1}{36\pi \times 10^9}$.

Substituting these values into the ratio:

Ratio $= \frac{10}{\frac{1}{36\pi \times 10^9} \times 2\pi \times 10^8} = \frac{10 \times 36\pi \times 10^9}{2\pi \times 10^8}$

Ratio $= 180 \times 10^1 = 1800$

Therefore, the ratio of maximum conduction current density to maximum displacement current density is 1800 .

Hence, the correct answer is **1800**.

Question2

The electric field of a plane electromagnetic wave, travelling in an unknown nonmagnetic medium is given by,

$$E_y = 20 \sin\left(3 \times 10^6 x - 4.5 \times 10^{14} t\right) \text{V/m}$$

(where x, t and other values have S.I. units). The dielectric constant of the medium is ____

(speed of light in free space is 3×10^8 m/s)

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Answer: 4

Solution:

The general standard mathematical form for a plane wave traveling in the positive x -direction is:

$$\mathrm{y}(\mathrm{x}, \mathrm{t}) = \mathrm{A} \sin(\mathrm{kx} - \omega \mathrm{t})$$

Where :

- A is the amplitude (maximum displacement).
- k is the angular wave number, related to the wavelength (λ) by $\mathrm{k} = \frac{2\pi}{\lambda}$.
- ω is the angular frequency, related to the time period (T) by $\omega = \frac{2\pi}{T}$.

The speed of propagation of traveling wave is given as,

$$v = \frac{\omega}{k}$$

From Maxwell's equations, the speed of an electromagnetic wave in a vacuum (c) depends on the fundamental constants of free space:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Where μ_0 is the permeability of free space and ϵ_0 is the permittivity of free space.

When light travels through a material medium, its speed (v) changes depending on the magnetic and electrical properties of that specific medium :

$$v = \frac{1}{\sqrt{\mu \epsilon}}$$

Where μ is the absolute permeability and ϵ is the absolute permittivity of the medium.

Relative to a vacuum :

- $\mu = \mu_0 \mu_{\mathrm{r}}$ (where μ_{r} is relative permeability)
- $\epsilon = \epsilon_0 \epsilon_{\mathrm{r}}$ (where ϵ_{r} is relative permittivity, also called the dielectric constant, often denoted as K)

$$\mathrm{v} = \frac{1}{\sqrt{(\mu_0 \mu_{\mathrm{r}})(\epsilon_0 \epsilon_{\mathrm{r}})}}$$

$$\Rightarrow v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \cdot \frac{1}{\sqrt{\mu_r \epsilon_r}}$$

$$\Rightarrow v = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

As the medium is non-magnetic so its relative permeability is exactly 1 ($\mu_{\mathrm{r}} = 1$).

$$\mathrm{v} = \frac{\mathrm{c}}{\sqrt{1 \times \epsilon_{\mathrm{r}}}}$$

$$\Rightarrow \mathrm{v} = \frac{\mathrm{c}}{\sqrt{\epsilon_{\mathrm{r}}}}$$

$$\Rightarrow \sqrt{\epsilon_{\mathrm{r}}} = \frac{\mathrm{c}}{\mathrm{v}}$$

$$\Rightarrow \epsilon_{\mathrm{r}} = \left(\frac{\mathrm{c}}{\mathrm{v}}\right)^2$$

From the given electric field equation :

$$E_y = 20 \sin\left(3 \times 10^6 x - 4.5 \times 10^{14} t\right)$$

By comparing this to the standard form $\mathrm{E_y} = \mathrm{E_0} \sin(\mathrm{kx} - \omega \mathrm{t})$:

- $\mathrm{k} = 3 \times 10^6 \mathrm{rad/m}$
- $\omega = 4.5 \times 10^{14} \mathrm{rad/s}$

The speed of the wave in the medium (v) :

$\mathrm{v} = \frac{\omega}{\mathrm{k}}$

$\Rightarrow \mathrm{v} = \frac{4.5 \times 10^{14}}{3 \times 10^6}$

$\Rightarrow \mathrm{v} = 1.5 \times 10^8 \mathrm{~m/s}$

We are given $c = 3 \times 10^8 \mathrm{~m/s}$.

$\epsilon_{\mathrm{r}} = \left(\frac{\mathrm{c}}{\mathrm{v}}\right)^2$

$\Rightarrow \epsilon_{\mathrm{r}} = \left(\frac{3 \times 10^8}{1.5 \times 10^8}\right)^2$

$\Rightarrow \epsilon_{\mathrm{r}} = (2)^2 = 4$

Therefore, the dielectric constant of the medium is 4 .

Hence, the correct answer is 4 .

Question3

The equation of the electric field of an electromagnetic wave propagating through free space is given by : $E = \sqrt{377} \sin \left(6.27 \times 10^3 t - 2.09 \times 10^{-5} x\right) \mathrm{N/C}$

The average power of the electromagnetic wave is $\left(\frac{1}{\alpha}\right) \mathrm{W/m}^2$. The value of α is

$$\left(\text{ Take } \sqrt{\frac{\mu_0}{\varepsilon_o}} = 377 \text{ in SI units }\right)$$

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Answer: 2

Solution:

The standard wave equation for the electric field component of an electromagnetic wave is given as $\mathrm{E} = \mathrm{E}_0 \sin(\omega \mathrm{t} - \mathrm{kx})$, where E_0 is the amplitude of electric field, ω is the angular frequency and k is the wave number.

The give equation for the electric field of electromagnetic wave is
$= \sqrt{377} \sin\left(6.27 \times 10^3 t - 2.09 \times 10^{-5} \mathrm{x}\right) \mathrm{N/C}$

So, electric field amplitude is $\mathrm{E}_0 = \sqrt{377}\ \mathrm{N/C}$.

The total energy density (energy per unit volume) of an electromagnetic wave is the sum of the electric field energy density ($\mathrm{u_E}$) and magnetic field energy density ($\mathrm{u_B}$).

$$\mathrm{u} = \mathrm{u_E} + \mathrm{u_B} = \frac{1}{2}\epsilon_0 \mathrm{E}^2 + \frac{1}{2}\mu_0\ \mathrm{B}^2$$

In an electromagnetic wave traveling in a vacuum, the instantaneous electric and magnetic fields are related by the speed of light (c):

$$\mathrm{E} = \mathrm{cB} \Rightarrow\ \mathrm{B} = \frac{\mathrm{E}}{\mathrm{c}}$$

So, the energy density for magnetic field is,

$$u_B = \frac{1}{2}\mu_0\left(\frac{E}{c}\right)^2 = \frac{E^2}{2\mu_0 c^2}$$

Since $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow c^2 = \frac{1}{\mu_0 \epsilon_0} \Rightarrow \mu_0 c^2 = \frac{1}{\epsilon_0}$.

$$\mathrm{u_B} = \frac{1}{2}\epsilon_0 \mathrm{E}^2$$

Thus, the energy is equally shared: $\mathrm{u_E} = \mathrm{u_B}$.

The total instantaneous energy density is:

$$\mathrm{u} = \mathrm{u_E} + \mathrm{u_B} = \frac{1}{2}\epsilon_0 \mathrm{E}^2 + \frac{1}{2}\epsilon_0 \mathrm{E}^2 = \epsilon_0 \mathrm{E}^2$$

Since the electric field oscillates sinusoidally $(E = E_0 \sin \omega t)$, the average value of E^2 over one cycle is $\frac{E_0^2}{2}$.

$$\overline{\mathrm{u}} = \epsilon_0\left(\frac{\mathrm{E}_0^2}{2}\right) = \frac{1}{2}\epsilon_0 \mathrm{E}_0^2$$

Intensity is defined as the energy crossing a unit area per unit time. It is related to the average energy density by the speed of the wave c :

$$\mathrm{I} = \overline{\mathrm{u}} \times \mathrm{c}$$

$$\Rightarrow I = \left(\frac{1}{2}\epsilon_0 E_0^2\right) c$$

The characteristic impedance of free space, $\mathrm{Z}_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$ (approx 377Ω).

$$I = \frac{1}{2}\epsilon_0 E_0^2\left(\frac{1}{\sqrt{\mu_0 \epsilon_0}}\right)$$

$$\Rightarrow I = \frac{E_0^2}{2}\left(\frac{\epsilon_0}{\sqrt{\mu_0 \epsilon_0}}\right) = \frac{E_0^2}{2}\sqrt{\frac{\epsilon_0}{\mu_0}} = \frac{(\sqrt{377})^2}{2} \times \frac{1}{377} = \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} = \frac{1}{\alpha} \Rightarrow \alpha = 2$$

Therefore, the value of α is 2.

Question4

The electric field in a plane electromagnetic wave is given by :

$$E_y = 69 \sin \left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right] \mathrm{V/m}.$$

The expression for magnetic field associated with this electromagnetic wave is ____ T.

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Options:

A.

$B_z = 2.3 \times 10^{-7} \sin \left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

B.

$B_z = 2.3 \times 10^{-7} \sin \left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

C.

$B_y = 2.3 \times 10^{-7} \sin \left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

D.

$B_y = 69 \sin \left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

Answer: A

Solution:

In a vacuum or air, the amplitudes of the electric field (E_0) and magnetic field (B_0) are related by the speed of light (c) :

$c = \frac{E_0}{B_0} \Rightarrow B_0 = \frac{E_0}{c}$

From the given equation $\mathrm{E_y} = 69 \sin \left[0.6 \times 10^3 \mathrm{x} - 1.8 \times 10^{11} \mathrm{t}\right], \mathrm{E_0} = 69\ \mathrm{V/m}, \omega$ (angular frequency) $= 1.8 \times 10^{11} \mathrm{rad/s}$ and k (wave number) $= 0.6 \times 10^3 \mathrm{rad/m}$

So, the amplitude of the magnetic field is,

$B_0 = \frac{69}{3\times10^8} = 23 \times 10^{-8} = 2.3 \times 10^{-7} T$

In electromagnetic waves which is a transverse waves consists of the electric field ($\overrightarrow{\mathrm{E}}$), magnetic field ($\overrightarrow{\mathrm{B}}$), and the direction of propagation ($\overrightarrow{\mathrm{s}}$) are all mutually perpendicular.

The wave is traveling along the +x direction and the electric field is oscillating along the y -axis.

Using the direction vector $\hat{\mathrm{s}} = \widehat{\mathrm{E}} \times \widehat{\mathrm{B}}$:

$\hat{\imath} = \hat{\jmath} \times \widehat{B}$

For this to be true, the magnetic field must be along the $\mathrm{z-axis}(\hat{\mathrm{k}})$, because $\hat{\mathrm{j}} \times \hat{\mathrm{k}} = \hat{\mathrm{i}}$.

In an electromagnetic wave in a vacuum, the electric and magnetic fields are in phase.

So, the phase of magnetic field vector is, $\left(0.6 \times 10^3\mathrm{x} - 1.8 \times 10^{11}\mathrm{t}\right)$

Therefore, the magnetic field vector is $\mathrm{B_z} = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3\mathrm{x} - 1.8 \times 10^{11}\mathrm{t}\right]$ T.

Hence, the correct option is (A).

Question5

A laser beam has intensity of $4.0 \times 10^{14}\ \mathrm{W/m^2}$. The amplitude of magnetic field associated with beam is ____ T.

(Take $\epsilon_o = 8.85 \times 10^{-12}\mathrm{C^2/Nm^2}$ and c $= 3 \times 10^8\ \mathrm{m/s}$)

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Options:

A.

1.83

B.

2.0

C.

5.5

D.

18.3

Answer: A

Solution:

The intensity I of an electromagnetic wave is related to the maximum electric field (E_0) and the maximum magnetic field (B_0). The formula for the intensity to the magnetic field amplitude is :

$I = \frac{cB_0^2}{2\mu_0}$

Where :

- I is the intensity.
- c is the speed of light in a vacuum.
- B_0 is the amplitude of the magnetic field.
- μ_0 is the permeability of free space.

The relationship between the speed of light (c), permeability (μ_0), and permittivity (ϵ_0) of free space is,

$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \Rightarrow \mu_0 = \frac{1}{\epsilon_0 c^2}$

By substituting μ_0 into the intensity equation, we get a formula directly relating the given variables :

$I = \frac{cB_0^2}{2\left(\frac{1}{\epsilon_0 c^2}\right)} = \frac{1}{2}c^3\epsilon_0 B_0^2$

The intensity of the laser beam is, $\mathrm{I} = 4.0 \times 10^{14}\ \mathrm{W/m^2}$

The permittivity of free space, $\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2/N \cdot m^2}$

The speed of light, $\mathrm{c} = 3 \times 10^8\ \mathrm{m/s}$

From the intensity formula, $B_0 = \sqrt{\frac{2I}{c^3\epsilon_0}}$

Putting the values,

$B_0 = \sqrt{\frac{2 \times 4.0 \times 10^{14}}{(3 \times 10^8)^3 \times 8.85 \times 10^{-12}}} = \sqrt{\frac{8.0 \times 10^{14}}{27 \times 10^{24} \times 8.85 \times 10^{-12}}}$

$\mathrm{B}_0 = \sqrt{\frac{8.0}{238.95} \times 10^2} \approx 1.83\ \mathrm{T}$

Therefore, the amplitude of the magnetic field associated with beam is approximately 1.83 T.

Hence, the correct option is (A).

Question6

Match List - I with List - II.

List - I Relation	List - II Law
A. $\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{d}{dt} \oint \vec{B} \cdot \overrightarrow{da}$	I. Ampere's circuital law
B. $\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt}\right)$	II. Faraday's laws of electromagnetic induction
C. $\oint \vec{E} \cdot \overrightarrow{da} = \frac{1}{\epsilon_0} \int_{\mathrm{v}} \rho \mathrm{dv}$	III. Ampere - Maxwell law
D. $\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 I$	IV. Gauss's law of electrostatics

Choose the correct answer from the options given below :

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Options:

A.

A-I, B-IV, C-III, D-II

B.

A-II, B-III, C-IV, D-I

C.

A-IV, B-I, C-II, D-III

D.

A-II, B-III, C-I, D-IV

Answer: B

Solution:

For relation A

The equation $\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt}[\oint \vec{B} \cdot d\vec{a}]$ mathematically states that the line integral of the electric field around a closed loop is equal to the negative rate of change of magnetic flux through the loop. This is the integral form of Faraday's Law of Electromagnetic Induction, i.e., A $\rightarrow$ II

For relation B

This equation $\oint \overrightarrow{\mathrm{B}} \cdot \overrightarrow{\mathrm{dl}}=\mu_0\left(\mathrm{I}+\epsilon_0 \frac{\mathrm{d} \Phi_{\mathrm{E}}}{\mathrm{dt}}\right)$ relates the magnetic field line integral to both the conduction current (I) and the displacement current $\left(\epsilon_0 \frac{\mathrm{d} \Phi_{\mathrm{E}}}{\mathrm{dt}}\right)$. This modification to Ampere's Law was introduced by Maxwell to account for changing electric fields. This is the Ampere-Maxwell Law, i.e., B $\rightarrow$ III

For relation C

In relation $C, \oint \vec{E} \cdot d \vec{a}=\frac{1}{\epsilon_0} \int_V \rho d V$ the total charge $Q_{\text {enclosed }}$ is represented as $\oint \vec{E} \cdot d \vec{a}$. The equation states that the total electric flux through a closed surface is equal to the enclosed charge divided by ϵ_0. This is Gauss's Law of Electrostatics, i.e., C $\rightarrow$ IV

For relation D

This relation $\oint \overrightarrow{\mathrm{B}} \cdot \overrightarrow{d \mathrm{l}}=\mu_0 \mathrm{I}$ states that the line integral of the magnetic field around a closed loop is equal to μ_0 times the current enclosed by the loop. This is the original form of Ampere's Circuital Law which is valid for steady currents, i.e., D $\rightarrow$ I

So, the correct match is A $\rightarrow$ II, B $\rightarrow$ III, C $\rightarrow$ IV, D $\rightarrow$ I. Hence, the correct option is (B).

Question7

The ratio of speeds of electromagnetic waves in vacuum and a medium, having dielectric constant $k=3$ and permeability of $\mu=2 \mu_0$, is ($\mu_0=$ permeability of vacuum)

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Options:

A.

6 : 1

B.

3 : 2

C.

$\sqrt{6}$: 1

D.

36 : 1

Answer: C

Solution:

The speed of light (or any electromagnetic wave) in a vacuum is given by the formula :

$c=\frac{1}{\sqrt{\mu_0 \epsilon_0}}$

Where :

- 1. μ_0 is the permeability of free space.
- 2. ϵ_0 is the permittivity of free space.

The speed of light in a specific medium is determined by that medium's permeability (μ) and permittivity (ϵ):

$v=\frac{1}{\sqrt{\mu \epsilon}}$

For the given medium :

Permeability : $\mu=2 \mu_0$

Dielectric Constant $\mathrm{k}=3$. The dielectric constant (relative permittivity ϵ_{r}) is defined as $\epsilon_{\mathrm{r}}=\frac{\epsilon}{\epsilon_0}$. Therefore, $\epsilon=\mathrm{k} \epsilon_0=3 \epsilon_0$.

We need to find the ratio $c: v$, which is the same as the refractive index n of the medium.

$\frac{c}{v}=\frac{1 / \sqrt{\mu_0 \epsilon_0}}{1 / \sqrt{\mu \epsilon}}$

$\Rightarrow \frac{c}{v}=\sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}=\sqrt{\frac{\left(2 \mu_0\right)\left(3 \epsilon_0\right)}{\mu_0 \epsilon_0}}$

For the medium $\mu=2 \mu_0$ and $\epsilon=3 \epsilon_0$:

$\frac{c}{v}=\sqrt{\frac{\left(2 \mu_0\right)\left(3 \epsilon_0\right)}{\mu_0 \epsilon_0}}=\frac{\sqrt{6}}{1}$

The ratio of the speed in vacuum to the speed in the medium is $\sqrt{6}: 1$.

Hence, the correct option is (C).

Question8

\text { Match the LIST-I with LIST-II }

List-I		List-II	
A.	Radio-wave	I.	is produced by Magnetron valve
B.	Micro-wave	II.	due to change in the vibrational modes of atoms
C.	Infrared-wave	III.	due to inner shell electrons moving from higher energy level to lower energy level
D.	X-ray	IV.	due to rapid acceleration of electrons

Choose the correct answer from the options given below:

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Options:

A.

A-IV, B-II, C-I, D-III

B.

A-IV, B-III, C-I, D-II

C.

A-IV, B-I, C-II, D-III

D.

A-II, B-IV, C-III, D-I

Answer: C

Solution:

Radio-waves

- 1. Radio waves have the longest wavelengths in the EM spectrum. They are artificially generated by transmitters. The core process involves an oscillating electric current in a special conductor called an antenna.
- 2. According to electromagnetic theory, an accelerating electric charge produces electromagnetic waves. In an antenna, electrons are rapidly accelerated back and forth.
- 3. So, the radio waves are produced due to rapid acceleration of electrons.

Micro-waves

- 1. Micro-waves are high-frequency radio waves. While they can be produced by acceleration, high-power microwaves (like those in radar or microwave ovens) are generated using special vacuum tubes.
- 2. The most common device for this is the Magnetron (or Klystron). A Magnetron uses a magnetic field to control the flow of electrons in a vacuum, causing them to oscillate at microwave frequencies.
- 3. Hence the Micro-wave is produced by Magnetron valve.

Infrared-waves

- 1. Infrared waves are often perceived as heat. All objects above absolute zero emit infrared radiation.

- 2. This emission arises from the thermal motion of atoms and molecules. When the atoms in a molecule vibrate or rotate, they change energy states. The transition between these vibrational and rotational energy levels results in the emission of infrared photons.

- 3. Hence, the infrared waves are produced due to change in the vibrational modes of atoms.

X-rays

- 1. X-rays have very high energy and short wavelengths. They are typically produced in X-ray tubes.

- 2. There are two main ways X-rays are produced:

1. (A) Braking Radiation: When high-speed electrons are suddenly stopped by a metal target.

2. (B) Characteristic X-rays: When a high-energy electron knocks an electron out of an inner shell (like the K-shell) of an atom, a vacancy is created. An electron from a higher energy outer shell falls into this vacancy. The large energy difference is released as an X-ray photon.

- 3. Hence, it is due to inner shell electrons moving from higher energy level to lower energy level.

So,

A (Radio-wave) → IV

B (Micro-wave) → I

C (Infrared-wave) → II

D (X-ray) → III

Hence, the correct option is (C).

Question9

The electric field of an electromagnetic wave travelling through a medium is given by $\vec{E}(x,t) = 25\sin\left(2.0 \times 10^{15}t - 10^{7}x\right)\hat{n}$ then the refractive index of the medium is ____ .

(All given measurement are in SI units)

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Options:

A.

2

B.

1.2

C.

1.5

D.

1.7

Answer: C

Solution:

The standard equation for a wave travelling in the +x direction is :

$\mathrm{E(x,t)} = \mathrm{E_0}\sin(\omega \mathrm{t} - \mathrm{kx})$

The given equation of electromagnetic wave is $\overrightarrow{\mathrm{E}}(\mathrm{x,t}) = 25\sin\left(2.0\times10^{15}\mathrm{t} - 10^7\mathrm{x}\right)\hat{\mathrm{n}}$:

Comparing both the equations,

Angular frequency $= \omega = 2.0\times10^{15}\mathrm{rad/s}$

Wave number $= \mathrm{k} = 10^7\ \mathrm{m}^{-1}$

The speed of the electromagnetic wave in the medium is given by the ratio of angular frequency to wave number:

$v = \frac{\omega}{k}$

$\mathrm{v} = \frac{2.0\times10^{15}}{10^7} = 2.0\times10^8\ \mathrm{m/s}$

The refractive index of a medium is the ratio of the speed of light in a vacuum (c) to the speed of light in that medium (v) :

$\mathrm{n} = \frac{\mathrm{c}}{\mathrm{v}}$

Taking the speed of light in vacuum as $\mathrm{c} = 3.0\times10^8\ \mathrm{m/s}$:

$\mathrm{n} = \frac{3.0\times10^8}{2.0\times10^8}$

$\mathrm{n} = 1.5$

Hence, the correct option is (C).

Question10

A plane electromagnetic wave is moving in free space with velocity $c = 3 \times 10^8$ m/s and its electric field is given as $\vec{E} = 54 \sin(kz - \omega t)\,\hat{j}$ V/m, where $\hat{j}$ is the unit vector along y-axis. The magnetic field vector $\vec{B}$ of the wave is :

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Options:

A.

$-1.8 \times 10^{-7} \sin(kz - \omega t)\,\hat{i}\ \mathrm{T}$

B.

$+1.8 \times 10^{-7} \sin(kz - \omega t)\,\hat{i}\ \mathrm{T}$

C.

$1.4 \times 10^{-7} \sin(kz - \omega t)\,\hat{k}\ \mathrm{T}$

D.

$1.4 \times 10^{-7} \sin(kz - \omega t)\,\hat{i}\ \mathrm{T}$

Answer: A

Solution:

In an electromagnetic wave traveling through a vacuum, the amplitudes of the electric field (E_0) and the magnetic field (B_0) are related by the speed of light (c):

$c = \frac{E_0}{B_0} \Rightarrow B_0 = \frac{E_0}{c}$

Given that $\mathrm{E}_0 = 54\ \mathrm{V/m}$ and $\mathrm{c} = 3 \times 10^8\ \mathrm{m/s}$

$B_0 = \frac{54}{3 \times 10^8}$

$\mathrm{B}_0 = 1.8 \times 10^{-7}\ \mathrm{T}$

In a transverse electromagnetic wave, the electric field (E), magnetic field (B), and the direction of propagation (v) are all mutually perpendicular. The direction follows the right-hand rule:

$\widehat{\mathrm{E}} \times \widehat{\mathrm{B}} = \widehat{\mathrm{v}}$

The variable z in phase expression ($kz - \omega t$) indicates the wave is moving along the $+z$ direction ($\hat{k}$).

The unit vector along direction of electric field ($\overrightarrow{\mathrm{E}}$) is given as $\hat{\mathrm{j}}$.

We need a unit vector such that $\hat{\mathrm{j}} \times (\widehat{\mathrm{B}}) = \hat{\mathrm{k}}$.

We know that $\hat{\jmath} \times \hat{\imath} = -\hat{\mathrm{k}}$ and $-\hat{\jmath} \times \hat{\imath} = \hat{\mathrm{k}}$

Therefore, if $\widehat{\mathrm{E}} = \hat{\mathrm{j}}$ and $\widehat{\mathrm{v}} = \widehat{\mathrm{k}}$, then $\widehat{\mathrm{B}}$ must be $-\hat{\mathrm{i}}$ because $\hat{\mathrm{j}} \times (-\mathrm{i}) = -(\hat{\mathrm{j}} \times \hat{\mathrm{i}}) = -(-\widehat{\mathrm{k}}) = \widehat{\mathrm{k}}$.

Combining the magnitude and the direction:

$\overrightarrow{\mathrm{B}} = -1.8 \times 10^{-7} \sin(\mathrm{kz} - \omega \mathrm{t})\hat{\mathrm{i}}\ \mathrm{T}$

Hence, the correct option is (A).

Question11

An electromagnetic wave travelling in x-direction is described by field equation

$E_y = 300 \sin \omega \left(t - \frac{x}{c}\right)$.

If the electron is restricted to move in y-direction only with speed of 1.5×10^6 m/s then ratio of maximum electric and magnetic forces acting on the electron is ______.

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Options:

A.

200

B.

150

C.

400

D.

Answer: A

Solution:

The electric force acting on a charge e is given by $\vec{F} = q\vec{E}$. The maximum force occurs when the electric field is at its peak (E_0):

$F_e(\max) = eE_0$

From the equation $E_y = 300 \sin \omega \left(t - \frac{x}{c}\right)$, the peak electric field is $E_0 = 300\ \mathrm{V/m}$.

The magnetic force on a charge is $\mathbf{F} = q(\vec{v} \times \vec{B})$, which has a maximum value of:

$F_m = evB_0$

Where B_0 is the peak magnetic field and v is the speed of the electron $(1.5 \times 10^6\ \mathrm{m/s})$.

In an electromagnetic wave, the amplitudes of the electric and magnetic fields are related by the speed of light (c):

$c = \frac{E_0}{B_0} \rightarrow B_0 = \frac{E_0}{c}$

The speed of light is $c \approx 3 \times 10^8\ \mathrm{m/s}$.

So, the ratio of maximum electric force to maximum magnetic force is:

$\frac{F_e}{F_m} = \frac{eE_0}{evB_0} = \frac{E_0}{vB_0}$

Substituting $B_0 = E_0/c$ into the ratio:

$\frac{F_e}{F_m} = \frac{E_0}{v(E_0/c)} = \frac{c}{v}$

$\frac{F_e}{F_m} = \frac{3 \times 10^8}{1.5 \times 10^6}$

$\frac{F_e}{F_m} = 2 \times 10^2 = 200$

Therefore, the ratio of maximum electric and magnetic forces is 200. Thus, the correct option is (A).

Question12

An electromagnetic wave travels in free space along the *x*-direction. At a particular point in space and time, $\vec{B} = 2 \times 10^{-7}\hat{j}$ T is associated with this wave. The value of corresponding electric field $\vec{E}$ at this point is _______ V/m.

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

$60\ \hat{k}$

B.

$-60\ \hat{k}$

C.

$30\ \hat{k}$

D.

$-600\ \hat{k}$

Answer: B

Solution:

Given:

- Wave travels along $+x$-direction
- Magnetic field is

  $\vec{B} = 2 \times 10^{-7}\,\hat{j}\ \mathrm{T}$

We know that for an electromagnetic wave,

$\vec{E} \times \vec{B}$

gives the direction of propagation.

Since the wave is moving along $+\hat{i}$ and $\vec{B}$ is along $+\hat{j}$, we need $\vec{E}$ such that

$\vec{E} \times \hat{j} = \hat{i}$

Now,

$(-\hat{k}) \times \hat{j} = \hat{i}$

So, $\vec{E}$ must be along $-\hat{k}$.

Now use the relation in free space:

$E = cB$

where

$c = 3 \times 10^8$ m/s

So,

$E = 3 \times 10^8 \times 2 \times 10^{-7}$

$E = 6 \times 10^1 = 60$ V/m

Therefore,

$\vec{E} = -60\,\hat{k}$ V/m

So, the correct answer is:

$\boxed{-60\,\hat{k} \text{ V/m}}$

Option B

Question13

A magnetic field vector in an electromagnetic wave is represented by $\vec{B} = B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{j}$. Its associated electric field vector is _____ .

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

$\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

B.

$\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{i}$

C.

$\vec{E} = v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

D.

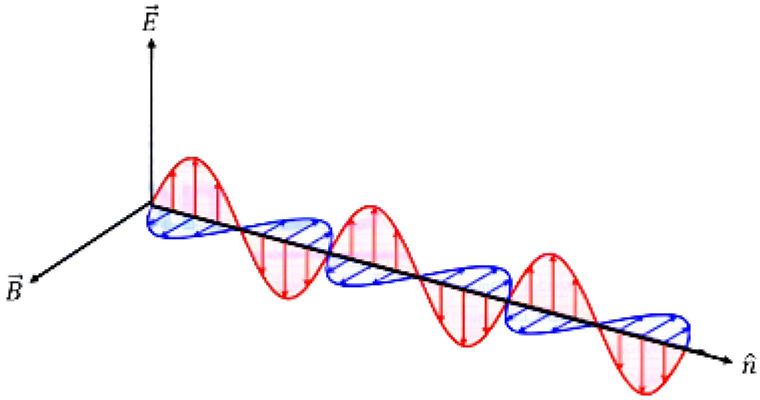
$\vec{E}=v\lambda B_0\sin\left(2\pi vt-\frac{2\pi x}{\lambda}\right)\hat{i}$

Answer: A

Solution:

The argument of the sine function is $\left(2\pi vt-\frac{2\pi x}{\lambda}\right)$. This indicates the wave is propagating along the positive x -axis.

Let the unit vector for the direction of propagation be $\hat{\mathrm{n}}=\hat{\mathrm{i}}$.



In an electromagnetic wave, the electric field vector $(\overrightarrow{\mathrm{E}})$, the magnetic field vector $(\overrightarrow{\mathrm{B}})$, and the direction of propagation ($\widehat{\mathrm{n}}$) are mutually perpendicular. Their relationship is given by:

$$\widehat{\mathrm{E}}=\widehat{\mathrm{B}}\times\widehat{\mathrm{n}}$$

It is given that $\overrightarrow{\mathrm{B}}$ is along $\hat{\mathrm{j}}$:

$$\widehat{\mathrm{E}}=\hat{\jmath}\times\hat{\imath}=-\hat{\mathrm{k}}$$

The magnitude of the electric field and magnetic field in a vacuum are related by the speed of light (c):

$$\mathrm{E}_0=\mathrm{cB}_0$$

The speed of light is the product of frequency and wavelength: $\mathrm{c}=\nu\lambda$.

$$\mathrm{E}_0=v\lambda\,\mathrm{B}_0$$

The electric field must have the same phase as the magnetic field. So, the electric field component is,

$$\overrightarrow{\mathrm{E}}=\mathrm{E}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi\mathrm{x}}{\lambda}\right)(-\widehat{\mathrm{k}})$$

$$\Rightarrow\overrightarrow{\mathrm{E}}=-v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi\mathrm{x}}{\lambda}\right)\hat{\mathrm{k}}$$

Therefore, the associated electric field vector is $-v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi\mathrm{x}}{\lambda}\right)\hat{\mathrm{k}}$.

Thus, the correct option is (A).

Question14

A displacement current of 4.0 A can be set up in the space between two parallel plates of $6 \mu \mathrm{~F}$ capacitor. The rate of change of potential difference across the plates of the capacitor is nearly $\alpha \times 10^6 \mathrm{~V} / \mathrm{s}$. The value of α is _____ .

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

0.58

B.

0.67

C.

0.82

D.

0.75

Answer: B

Solution:

According to Maxwell's law, the displacement current in a capacitor is equal to the rate of change of charge on the plates $\left(\mathrm{I}_{\mathrm{d}}=\frac{\mathrm{dQ}}{\mathrm{dt}}\right)$.

The charge (Q) on a capacitor of capacitance (C), connected to potential difference (V) is:

$Q=C V$

On differentiating both sides with respect to time (t) :

$\frac{d Q}{d t}=C \frac{d V}{d t}$

Since displacement current is $I_d=\frac{d Q}{d t}$, we have:

$I_d=C \frac{d V}{d t}$

The displacement current is given as $I_d = 4.0$ A and the capacitance of capacitor is $C = 6\mu\,F = 6 \times 10^{-6}$ F : Putting the values,

$$\frac{dV}{dt} = \frac{I_d}{C}$$

$$\Rightarrow \frac{dV}{dt} = \frac{4.0}{6\times10^{-6}}$$

$$\Rightarrow \frac{dV}{dt} = \frac{2}{3} \times 10^6 \approx 0.67 \times 10^6 \frac{V}{s} = \alpha \times 10^6 \frac{V}{s}$$

$$\Rightarrow \alpha = 0.67$$

Therefore, the correct option is (B).

Question15

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The electromagnetic wave exerts pressure on the surface on which they are allowed to fall.

Reason (R) : There is no mass associated with the electromagnetic waves.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

Both (A) and (R) are true and (R) is the correct explanation of (A)

B.

Both (A) and (R) are true but (R) is not the correct explanation of (A)

C.

(A) is true but (R) is false

D.

(A) is false but (R) is true

Answer: B

Solution:

Electromagnetic waves contain both energy (E) and momentum (p). And when an EM wave strikes a surface, it transfers its momentum to that surface.

According to Newton's Second Law, the rate of change of momentum is force, and force per unit area is pressure (radiation pressure).

Therefore, assertion (A) is true.

Electromagnetic waves consist of photons. Photons have zero rest mass. Though photons have energy and momentum, they do not have mass.

Therefore, reason (R) is true.

The reason EM waves exert pressure is because they carry momentum ($\mathrm{p}=\frac{\mathrm{E}}{\mathrm{c}}$), not because they don't have mass. If waves had no momentum, they wouldn't exert pressure regardless of their mass status. Thus, while reason is a true fact about EM waves, it does not explain why they exert pressure.

So, both (A) and (R) are true, but (R) is not the correct explanation of (A).

Thus, the correct option is (B).

Question16

A point light source emits E.M. waves in free space. A detector, placed at a distance of L m, measures the intensity as I_{o}. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be ____ .

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

$\frac{I_{\mathrm{o}}}{4}$

B.

I_o

C.

$\frac{I_0}{\sqrt{2}}$

D.

$\frac{I_o}{2}$

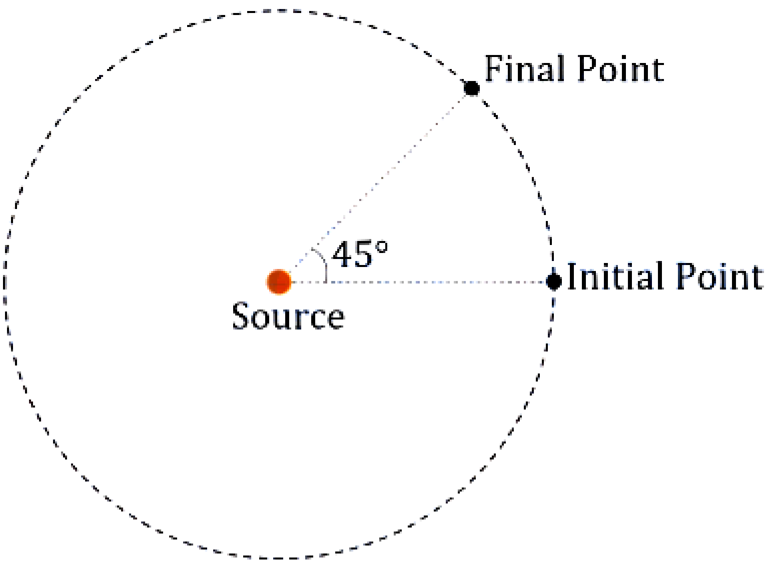
Answer: B

Solution:

The intensity (I) of electromagnetic waves from a point source in free space follows the inverse square law.

$I = \frac{P}{4\pi r^2}$

Where P is the power of the source and r is the distance from the source.



Initial distance is L and the intensity recorded is I_0.

It is given that the detector is shifted to another location on the same spherical surface.

On a spherical surface is all points are at a constant distance r from the center (the source).

Since the detector remains on the same spherical surface, its distance from the point source remains L .

The angle (45°) represents a change in direction, but not a change in distance r .

As r is constant $\mathrm{r}_1 = \mathrm{r}_2 = \mathrm{L}$:

$I_{\text{new}} = \frac{P}{4\pi L^2} = I_0$

Therefore, the measured intensity at new location will be I_0. Hence, the correct option is (B).

Question17

For an electromagnetic wave propagating through vacuum, $\vec{k}$, $\vec{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by:

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

$\frac{\vec{E}\times\vec{k}}{\omega}$

B.

$\frac{\vec{k}\times\vec{E}}{\omega}$

C.

$\omega(\vec{E}\times\vec{k})$

D.

$\omega(\vec{k}\times\vec{E})$

Answer: B

Solution:

In an electromagnetic wave, the electric field $\vec{E}$, magnetic field $\vec{B}$, and the propagation direction $\vec{k}$ are mutually perpendicular to each other. The wave travels in the direction of the cross product $\overrightarrow{\mathrm{E}} \times \overrightarrow{\mathrm{B}}$.

Also the magnetic field vector is in the direction of $\widehat{\mathrm{k}} \times \widehat{\mathrm{E}}$

The magnitude of magnetic field is given as $\mathrm{B} = \frac{\mathrm{E}}{\mathrm{c}}$, where c is the speed of light.

Maxwell's equation (Faraday's Law) in differential form is :

$\nabla \times \overrightarrow{\mathrm{E}} = -\frac{\partial\overrightarrow{\mathrm{B}}}{\partial \mathrm{t}}$

For a plane monochromatic wave, the fields can be represented asb :

$\overrightarrow{\mathrm{E}}=\overrightarrow{\mathrm{E}}_0 \mathrm{e}^{\mathrm{i}(\overrightarrow{\mathrm{k}} \cdot \overrightarrow{\mathrm{r}}-\omega \mathrm{t})}$

$\overrightarrow{\mathrm{B}}=\overrightarrow{\mathrm{B}}_0 \mathrm{e}^{\mathrm{i}(\overrightarrow{\mathrm{k}} \cdot \overrightarrow{\mathrm{r}}-\omega \mathrm{t})}$

Applying the curl operator ($\nabla \times$) to the electric field expression :

$\nabla \times \overrightarrow{\mathrm{E}}=\mathrm{i} \overrightarrow{\mathrm{k}} \times \overrightarrow{\mathrm{E}}$

Applying the time derivative to the magnetic field expression :

$\frac{\partial \overrightarrow{\mathrm{B}}}{\partial \mathrm{t}}=-\mathrm{i} \omega \overrightarrow{\mathrm{B}}$

Substituting into Faraday's Law :

$\mathrm{i} \overrightarrow{\mathrm{k}} \times \overrightarrow{\mathrm{E}}=-(-\mathrm{i} \omega \overrightarrow{\mathrm{B}})$

$\Rightarrow \mathrm{i} \overrightarrow{\mathrm{k}} \times \overrightarrow{\mathrm{E}}=\mathrm{i} \omega \overrightarrow{\mathrm{B}}$

$\Rightarrow \overrightarrow{\mathrm{B}}=\frac{\overrightarrow{\mathrm{k}} \times \overrightarrow{\mathrm{E}}}{\omega}$

Therefore, the correct option is (B)

Question18

A parallel plate capacitor of area $A=16 \mathrm{~cm}^2$ and separation between the plates 10 cm , is charged by a DC current. Consider a hypothetical plane surface of area $\mathrm{A}_0=3.2 \mathrm{~cm}^2$ inside the capacitor and parallel to the plates. At an instant, the current through the circuit is 6A. At the same instant the displacement current through A_0 is ___________ mA .

JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 1200

Solution:

To determine the displacement current through a hypothetical plane surface within a parallel plate capacitor, follow these steps:

Current Density Calculation:

The current density (J_d) is the current (I) divided by the area (A) of the capacitor plates.

Given: $I = 6\,\mathrm{A}$ and $A = 16\,\mathrm{cm}^2$.

$J_d = \frac{I}{A} = \frac{6\,\mathrm{A}}{16\,\mathrm{cm}^{2}}$

Displacement Current Through the Hypothetical Surface:

The hypothetical surface has an area $A_0 = 3.2\,\mathrm{cm}^2$.

The displacement current through this smaller area (I_{small}) is the current density multiplied by the area of the hypothetical surface.

$I_{small} = J_d \times A_0 = \frac{6\,\mathrm{A}}{16\,\mathrm{cm}^{2}} \times 3.2\,\mathrm{cm}^2$

Calculation:

Simplify the expression to find the displacement current:

$I_{small} = \left(\frac{6}{16}\right) \times 3.2 = 1.2\,\mathrm{A} = 1200\,\mathrm{mA}$

Thus, the displacement current through the hypothetical plane surface is 1200 mA.

Question19

A time varying potential difference is applied between the plates of a parallel plate capacitor of capacitance 2.5μ F. The dielectric constant of the medium between the capacitor plates is 1 . It produces an instantaneous displacement current of 0.25 mA in the intervening space between the capacitor plates, the magnitude of the rate of change of the potential difference will be _________ Vs^{-1}.

JEE Main 2025 (Online) 23rd January Evening Shift

Answer: 100

Solution:

$\frac{dV}{dt} = \frac{I}{C}$

Given that the displacement current is $I = 0.25 \times 10^{-3}$ A and the capacitance is $C = 2.5 \times 10^{-6}$ F, the rate of change of the potential difference is calculated as follows:

$\frac{dV}{dt} = \frac{0.25 \times 10^{-3}}{2.5 \times 10^{-6}} = \frac{0.25}{2.5} \times \frac{10^{-3}}{10^{-6}} = 0.1 \times 10^{3} = 100\,\text{V/s}.$

Thus, the magnitude of the rate of change of the potential difference is 100 V/s.

Question20

A plane electromagnetic wave propagating in x-direction is described by

$E_y = (200\text{Vm}^{-1}) \sin [1.5 \times 10^{7} t - 0.05x]$;

The intensity of the wave is :

(Use $E_0 = 8.85 \times 10^{-12} \text{C}^2\text{N}^{-1}\text{m}^{-2}$)

[27-Jan-2024 Shift 1]

Options:

A.

35.4Wm^{-2}

B.

53.1Wm^{-2}

C.

26.6Wm^{-2}

D.

106.2Wm^{-2}

Answer: B

Solution:

$$I = \frac{1}{2}\varepsilon_0 {E_0}^2 \times c$$

$$I = \frac{1}{2} \times 8.85 \times 10^{-12} \times 4 \times 10^4 \times 3 \times 10^8$$

$$I = 53.1 \mathrm{W/m^2}$$

Question21

An object is placed in a medium of refractive index 3. An electromagnetic wave of intensity $6 \times 10^8 \mathrm{W/m^2}$ falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space $= 3 \times 10^8 \mathrm{m/s}$):

[27-Jan-2024 Shift 2]

Options:

A.

$36 \mathrm{Nm^{-2}}$

B.

$18 \mathrm{Nm^{-2}}$

C.

$6 \mathrm{Nm^{-2}}$

D.

$2 \mathrm{Nm^{-2}}$

Answer: C

Solution:

Radiation pressure $= \frac{I}{V}$

$= \frac{I \cdot \mu}{c}$

$= \frac{6 \times 10^8 \times 3}{3 \times 10^8}$

$= 6 N / m^2$

Question22

A plane electromagnetic wave of frequency 35MHz travels in free space along the X-direction. At a particular point (in space and time) $\vec{E} = 9.6 \hat{j} V / m$. The value of magnetic field at this point is :

[29-Jan-2024 Shift 2]

Options:

A.

$3.2 \times 10^{-8} \hat{k} T$

B.

$3.2 \times 10^{-8} \hat{i} T$

C.

$9.6 \hat{j} T$

D.

$9.6 \times 10^{-8} \hat{k} T$

Answer: A

Solution:

$\frac{E}{B} = C$

$\frac{E}{B} = 3 \times 10^8$

$B = \frac{E}{3 \times 10^8} = \frac{9.6}{3 \times 10^8}$

$B = 3.2 \times 10^{-8} \mathrm{T}$

$\hat{B} = \hat{v} \times \hat{E}$

$= \hat{i} \times \hat{j} = \hat{k}$

So,

$\vec{B} = 3.2 \times 10^{-8} \hat{k} \mathrm{T}$

Question23

The electric field of an electromagnetic wave in free space is represented as $\vec{E} = E_0 \cos(\omega t - kz) \hat{i}$.

The corresponding magnetic induction vector will be :

[30-Jan-2024 Shift 1]

Options:

A.

$\vec{B} = E_0 C \cos(\omega t - kz) \hat{j}$

B.

$\vec{B} = \frac{E_0}{C} \cos(\omega t - kz) \hat{j}$

C.

$\vec{B} = E_0 \mathrm{Cos}(\omega t + kz) \hat{j}$

D.

$$\vec{B} = \frac{E_0}{C}\cos(\omega t + kz)\hat{j}$$

Answer: B

Solution:

Given $\vec{E} = E_0 \cos(\omega t - kz)\hat{i}$

$$\vec{B} = \frac{E_0}{C}\cos(\omega t - kz)\hat{j}$$

$$\hat{C} = \hat{E} \times \hat{B}$$

Question24

Match List I with List II

	List-I		List-II
A.	Gauss's law of magnetostatics	I.	$\oint \vec{E} \cdot \vec{d}a = \frac{1}{\varepsilon_0}\int \rho \, dV$
B.	Faraday's law of electro magnetic induction	II.	$\oint \vec{B} \cdot \vec{d}a = -0$
C.	Ampere's law	III.	$\oint \vec{E} \cdot \vec{d}l = \frac{-d}{dt}\int \vec{B} \cdot \vec{d}a$
D.	Gauss's law of electrostatics	IV.	$\oint \vec{B} \cdot \vec{d}l = -\mu_0 I$

Choose the correct answer from the options given below:

[30-Jan-2024 Shift 2]

Options:

A.

A-I, B-III, C-IV, D-II

B.

A-III, B-IV, C-I, D-II

C.

A-IV, B-II, C-III, D-I

D.

A-II, B-III, C-IV, D-I

Answer: D

Solution:

Maxwell's equation

Question25

In a plane EM wave, the electric field oscillates sinusoidally at a frequency of 5×10^{10}Hz and an amplitude of $50 Vm^{-1}$. The total average energy density of the electromagnetic field of the wave is : [Use $\varepsilon 0 = 8.85 \times 10^{-12} C^2/Nm^2$]

[31-Jan-2024 Shift 1]

Options:

A.

$1.106 \times 10^{-8} Jm^{-3}$

B.

$4.425 \times 10^{-8} Jm^{-3}$

C.

$2.212 \times 10^{-8} Jm^{-3}$

D.

$2.212 \times 10^{-10} Jm^{-3}$

Answer: A

Solution:

$$U_E = \frac{1}{2}\varepsilon_0 E^2$$

$$U_E = \frac{1}{2} \times 8.85 \times 10^{-12} \times (50)^2$$

$$= 1.106 \times 10^{-8} J/m^3$$

Question26

Given below are two statements:

Statement I: Electromagnetic waves carry energy as they travel through space and this energy is equally shared by the electric and magnetic fields.
Statement II: When electromagnetic waves strike a surface, a pressure is exerted on the surface.

In the light of the above statements, choose the most appropriate answer from the options given below:

[31-Jan-2024 Shift 2]

Options:

A.

Statement I is incorrect but Statement II is correct

B.

Both Statement I and Statement II are correct.

C.

Both Statement I and Statement II are incorrect.

D.

Statement I is correct but Statement II is incorrect.

Answer: B

Solution:

$$\frac{1}{2}\varepsilon_0 E^2 = \frac{B^2}{2\mu_0}$$

$$\because E = CB \text{ and } C = \frac{1}{\mu_0 \varepsilon_0}$$

Question27

If frequency of electromagnetic wave is 60MHz and it travels in air along z direction then the corresponding electric and magnetic field vectors will be mutually perpendicular to each other and the wavelength of the wave (in m) is :

[1-Feb-2024 Shift 2]

Options:

A.

2.5

B.

10

C.

5

D.

2

Answer: C

Solution:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{60 \times 10^6} = 5\text{m}$$

Question28

If $\vec{E}$ and $\vec{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by : ($\omega-$ angular frequency) :
[24-Jan-2023 Shift 1]

Options:

A. $\frac{1}{\omega}\left(\vec{K} \times \vec{E}\right)$

B. $\omega\left(\vec{E} \times \vec{K}\right)$

C. $\omega\left(\vec{K} \times \vec{E}\right)$

D. $\vec{K} \times \vec{E}$

Answer: A

Solution:

Magnetic field vector will be in the direction of $\hat{K} \times \hat{E}$
magnitude of $B = \frac{E}{C} = \frac{K}{\omega}E$
Or $\vec{B} = \frac{1}{\omega}\left(\vec{K} \times \vec{E}\right)$

Question29

The electric field and magnetic field components of an electromagnetic wave going through vacuum is described by
$E_x = E_0 \sin(kz - \omega t)$
$B_y = B_0 \sin(kz - \omega t)$
Then the correct relation between E_0 and B_0 is given by
[24-Jan-2023 Shift 2]

Options:

A. $kE_0 = \omega B_0$

B. $E_0 B_0 = \omega k$

C. $\omega E_0 = kB_0$

D. $E_0 = kB_0$

Answer: A

Solution:

$$C = \frac{\omega}{k} = \frac{E_0}{B_0}$$

Question30

All electromagnetic wave is transporting energy in the negative z direction. At a certain point and certain time the direction of electric field of the wave is along positive y direction. What will be the direction of the magnetic field of the wave at that point and instant?
[25-Jan-2023 Shift 1]

Options:

A. Positive direction of x

B. Positive direction of z

C. Negative direction of x

D. Negative direction of y

Answer: A

Solution:

As , poynting vector
$\vec{S} = \vec{E} \times \vec{H}$
Given energy transport = negative z direction
Electric field = positive y direction
$(-\hat{k}) = (+\hat{j}) \times [\hat{i}]$
Hence according to vector cross product magnetic field should be positive x direction.

Question31

Which of the following are true?
A. Speed of light in vacuum is dependent on the direction of propagation.
B. Speed of light in a medium in independent of the wavelength of light.
C. The speed of light is independent of the motion of the source.
D. The speed of light in a medium is independent of intensity.
Choose the correct answer from the option given below :
[29-Jan-2023 Shift 1]

Options:

A. A and C only

B. B and D only

C. B and C only

D. C and D only

Answer: D

Solution:

Speed of light does not depend on the motion of source as well as intensity.

Question32

Given below are two statements:
Statement I : Electromagnetic waves are not deflected by electric and magnetic field.
Statement II : The amplitude of electric field and the magnetic field in electromagnetic waves are related to each other as $E_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} B_0$

In the light of the above statements, choose the correct answer from the options given below:
[29-Jan-2023 Shift 2]

Options:

A. Statement I is true but statement II is false

B. Both Statement I and Statement II are true

C. Statement I is false but statement II is true

D. Both Statement I and Statement II are false

Answer: A

Solution:

Statement-I is correct as EMW are neutral.
Statement-II is wrong.

$E_0 = \sqrt{\frac{1}{\mu_0 \varepsilon_0}} B_0$

Question33

Match List-I with List-II.

List-I	List-II
A. Microwaves	I. Physiotherapy
B. UV rays	II. Treatment of cancer
C. Infra-red rays	III.Lasik eye surgery
D. X-rays	IV. Aircraft navigation

Choose the correct answer from the option given below: [31-Jan-2023 Shift 2]

Options:

A. A-II, B-IV, C-III, D-I

B. A-IV, B-I, C-II, D-III

C. A-IV, B-III, C-I, D-II

D. A-III, B-II, C-I, D-IV

Answer: C

Question34

Match the List-I with List-II:

List I	List II
A. Microwaves	I. Radio active decay of the nucleus
B. Gamma rays	II. Rapid acceleration and deceleration of electron in aerials
C. Radio waves	III. Inner shell electrons
D. X-rays	IV. Klystron valve

Choose the correct answer from the options given below:
[1-Feb-2023 Shift 1]

Options:

A. A-I, B-II, C-III, D-IV

B. A-IV, B-I, C-II, D-III

C. A-I, B-III, C-IV, D-II

D. A-IV, B-III, C-II, D-I

Answer: B

Solution:

Based on theory.

Question35

The ratio of average electric energy density and total average energy density of electromagnetic wave is:
[1-Feb-2023 Shift 2]

Options:

A. 2

B. 1

C. 3

D. $\frac{1}{2}$

Answer: D

Solution:

$u_E > = u_B > = \frac{1}{2} < u_{total} > ..$

So $\frac{u_E \rangle}{u_{total} \rangle} = \frac{1}{2}$

Question36

The energy density associated with electric field $\bar{E}$ and magnetic field $\bar{B}$ of an electromagnetic wave in free space is given by (ε_0 – . permittivity of free space, μ_0 - permeability of free space) [6-Apr-2023 shift 2]

Options:

A. $U_E = \frac{\epsilon_0 E^2}{2}, U_B = \frac{B^2}{2\mu_0}$

B. $U_E = \frac{E^2}{2E_0}, U_B = \frac{\mu_0 B^2}{2}$

C. $U_E = \frac{E^2}{2E_0}, U_B = \frac{B^2}{2\mu_0}$

D. $U_E = \frac{\epsilon_0 E^2}{2}, U_B = \frac{\mu_0 B^2}{2}$

Answer: A

Solution:

By theory of electromagnetic waves

$U_E = \frac{1}{2}\varepsilon_0 E^2$ and

$U_B = \frac{1}{2}\frac{B^2}{\mu_0}$

Question37

The waves emitted when a metal target is bombarded with high energy electrons are

[8-Apr-2023 shift 2]

Options:

A. Microwaves

B. X-rays

C. Radio Wa̅ves

D. Infrared rays

Answer: B

Solution:

By theory

Question38

The amplitude of magnetic field in an electromagnetic wave propagating along y-axis is 6.0×10^{-7}T. The maximum value of electric field in the electromagnetic wave is [10-Apr-2023 shift 2]

Options:

A. $2 \times 10^{15} Vm^{-1}$

B. $2 \times 10^{14} Vm^{-1}$

C. $6.0 \times 10^{-7} Vm^{-1}$

D. $180 Vm^{-1}$

Answer: D

Solution:

In electromagnetic wave, $E_0 = B_0 C$

$E_0 = 6 \times 10^{-7} \times 3 \times 10^{8}$ $E_0 \rightarrow$ Amplitude of electric field

$= 18 \times 10^{1}$ $B_0 \rightarrow$ Amplitude of magnetic field
$= 180 v/m$ $C \rightarrow$ Speed of light

Question39

A plane electromagnetic wave of frequency 20 MHz propagates in free space along x-direction. At a particular space and time, $\vec{E} = 6.6\hat{j} v/m$. What is $\vec{B}$ at this point? [11-Apr-2023 shift 2]

Options:

A. $-2.2 \times 10^{-8}\hat{k} T$

B. $-2.2 \times 10^{-8}\hat{i} T$

C. $2.2 \times 10^{-8}\hat{k} T$

D. $2.2 \times 10^{-8}\hat{i} T$

Answer: C

Solution:

$|B| = \frac{|E|}{C} = \frac{6.6}{3 \times 10^{8}} = 2.2 \times 10^{-8}$

For direction of $\vec{B}$

$= \vec{E} \times \vec{B} = \vec{C}$

$= \hat{j} \times \vec{B} = \hat{i}$

$\vec{B} = (2.2 \times 10^{-8})\hat{k} T$

Question40

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : EM waves used for optical communication have longer wavelength than that of microwave, employed in Radar technology.

Reason R : Infrared EM waves are more energetic than microwaves, (used in Radar) In the light of given statements, choose the correct answer from the options given below.
[12-Apr-2023 shift 1]

Options:

A. A is true but R is false

B. Both A and R are true and r is the correct explanation of A

C. A is false but R is true

D. Both A and R are true but R is NOT the correct explanation of A

Answer: C

Question41

Which of the following Maxwell's equation is valid for time varying conditions but not valid for static conditions:
[13-Apr-2023 shift 1]

Options:

A. $\oint \vec{D} \cdot \vec{dA} = Q$

B. $\oint \vec{E} \cdot \vec{dl} = -\frac{\partial \varphi_B}{\partial t}$

C. $\oint \vec{E} \cdot \vec{dl} = 0$

D. $\oint \vec{B} \cdot \vec{dl} = \mu_0 I$

Answer: B

Solution:

For static conditions

$$\oint \vec{E} \cdot d\vec{l} = 0$$

For time varying condition,

$$\oint \vec{E} \cdot d\vec{l} = -\frac{\partial \varphi_B}{\partial t}$$

Question42

Given below are two statements :
Statement I : Out of microwaves, infrared rays and ultraviolet rays, ultraviolet rays are the most effective for the emission of electrons from a metallic surface.
Statement II : Above the threshold frequency, the maximum kinetic energy of photoelectrons is inversely proportional to the frequency of the incident light.
In the light of above statements, choose the correct answer form the options given below
[13-Apr-2023 shift 2]

Options:

A. Statement I is false but statement II is true

B. Both Statement I and Statement II are true

C. Statement I is true but statement II is false

D. Both Statement I and Statement II are false

Answer: C

Solution:

UV rays have maximum frequency hence are most effective for emission of electrons from the metallic surface.

$$KE_{max} = hf - hf_0$$

Question43

In an electromagnetic wave, at an instant and at a particular position, the electric field is along the negative z axis and magnetic field is along the positive x-axis. Then the direction of propagation of electromagnetic wave is :
[13-Apr-2023 shift 2]

Options:

A. negative y-axis

B. at 45° angle from positive y-axis

C. positive y-axis

D. positive z-axis

Answer: A

Solution:

Direction of propagation of EM wave will be in the direction of $\left(\vec{E} \times \vec{B}\right)$

Question44

Match List I with List II of Electromagnetic waves with corresponding wavelength range :

List I	List II
(A) Microwave	(I) 400 nm to 1 nm
(B) Ultraviolet	(II) 1 nm to 10^{-3} nm
(C) X-Ray	(III) 1 mm to 700 nm
(D) Infra-rad	(IV) 0.1m to 1 mm

Choose the correct answer from the options given below :
[15-Apr-2023 shift 1]

Options:

A. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

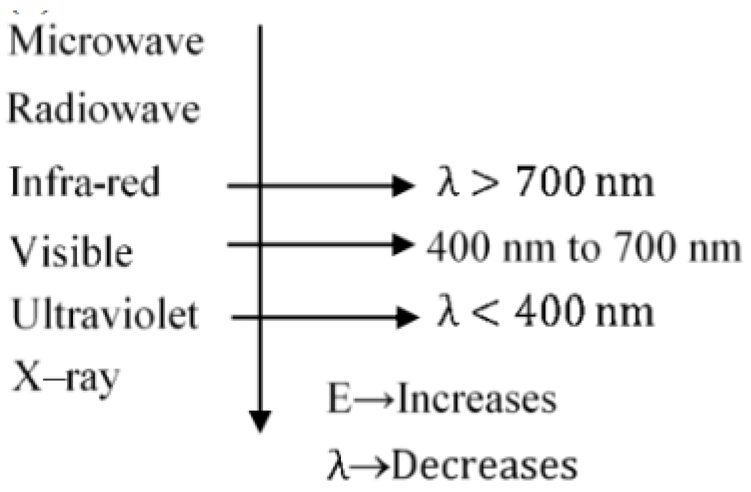
B. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

C. (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

D. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Answer: B

Solution:



Question45

A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44. If magnitude of magnetic intensity is $4.5 \times 10^{-2} Am^{-1}$ at a point, what will be the approximate magnitude of electric field intensity at that point? (Given : Permeability of free space $\mu_0 = 4\pi \times 10^{-7} NA^{-2}$, speed of light in vacuum $c = 3 \times 10^8 ms^{-}1$)
[24-Jun-2022-Shift-1]

Options:

A. $16.96 Vm^{-1}$

B. $2.25 \times 10^{-2} Vm^{-1}$

C. $8.48 Vm^{-1}$

D. $6.75 \times 10^{6} Vm^{-1}$

Answer: C

Solution:

$H = 4.5 \times 10^{-2}$
So $B = \mu_0 \mu H$

Thus $E = \frac{c}{n} B$ (where $n \Rightarrow$ refractive index)

So $E = \frac{3 \times 10^{8} \times 4\pi \times 10^{-7} \times 1.61 \times 4.5 \times 10^{-2}}{\sqrt{1.61 \times 6.44}}$

$E = 8.48$

Question46

The electric field in an electromagnetic wave is given by $E = 56.5 \sin \omega(t - x / c) NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space. (Given : $\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$) [25-Jun-2022-Shift-1]

Options:

A. $5.65 Wm^{-2}$

B. $4.24 Wm^{-2}$

C. $1.9 \times 10^{-7} Wm^{-2}$

D. $56.5 Wm^{-2}$

Answer: B

Solution:

$$I = \frac{1}{2}\varepsilon_0 E_0^{\ 2} c$$
$$= \frac{1}{2} 8.5 \times 10^{-12} \times (56.5)^2 \times 3 \times 10^8$$
$$= 4.24 W/m^2$$

Question47

The electromagnetic waves travel in a medium at a speed of $2.0 \times 10^8 m/s$. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be :
[25-Jun-2022-Shift-2]

Options:

A. 2.25

B. 4.25

C. 6.25

D. 8.25

Answer: A

Solution:

$$n = \frac{c}{v} = \frac{3}{2}$$
$$\sqrt{\in\mu} = n$$
$$\text{So } \in = \frac{9}{4} = 2.25$$

Question48

If Electric field intensity of a uniform plane electromagnetic wave is given as $E = -301.6 \sin(kz - \omega t)\hat{a}_x + 452.4 \sin(kz - \omega t)\hat{a}_y \frac{V}{m}$. Then magnetic intensity ' H' of this wave in Am^{-1} will be:
[Given : Speed of light in vacuum $c = 3 \times 10^8 ms^{-1}$, Permeability of

vacuum .$\mu_0 = 4\pi \times 10^{-7} NA^{-2}$]
[26-Jun-2022-Shift-1]

Options:

A. $+0.8 \sin(kz - \omega t)\,\hat{a}_y + 0.8 \sin(kz - \omega t)\,\hat{a}_x$

B. $+1.0 \times 10^{-6} \sin(kz - \omega t)\,\hat{a}_y + 1.5 \times 10^{-6}(kz - \omega t)\hat{a}_x$

C. $-0.8 \sin(kz - \omega t)\,\hat{a}_y - 1.2 \sin(kz - \omega t)\,\hat{a}_x$

D. $-1.0 \times 10^{-6} \sin(kz - \omega t)\,\hat{a}_y - 1.5 \times 10^{-6} \sin(kz - \omega t)\,\hat{a}_x$

Answer: C

Solution:

$\vec{E} = -301.6 \sin(kz - \omega t)\,\hat{a}_x + 452.4 \sin(kz - \omega t)\,\hat{a}_y$
$E_{0x} = 301.6$
$E_{0y} = +452.4$
$E_0 = \sqrt{E_{0x}^{\ 2} + E_{0y}^{\ 2}}$

Now, $\frac{E_0}{B_0} = C \Rightarrow B_0 = \frac{E_0}{C} = \frac{\sqrt{E_{0x}^{\ 2} + E_{0y}^{\ 2}}}{C}$

Also, $\hat{B} = \hat{C} \times \hat{E} \Rightarrow \hat{k} \times \frac{\left(E_{0x}\hat{i} + E_{0y}\hat{j}\right)}{\sqrt{E_{0x}^{\ 2} + E_{0y}^{\ 2}}}$

$\hat{B} = \frac{-E_{0x}\hat{j} - E_{0y}\hat{i}}{\sqrt{E_{0x}^{\ 2} + E_{0y}^{\ 2}}}$

$\vec{B} = -\frac{E_{0x}}{C} \sin(kz - \omega t)\,\hat{a}_y - \frac{E_{0y}}{C} \sin(kz - \omega t)\,\hat{a}_x$

$\vec{H} = \frac{\vec{B}}{\mu_0}$

$\vec{H} = -\frac{E_{0x}}{\mu_0 C} \sin(kz - \omega t)\,\hat{a}_y - \frac{E_{0y}}{\mu_0 C} \sin(kz - \omega t)\,\hat{a}_x$

Question49

Which is the correct ascending order of wavelengths? [26-Jun-2022-Shift-2]

Options:

A. $\lambda_{\text{visible}} < \lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{microwave}}$

B. $\lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

C. $\lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

D. $\lambda_{\text{microwave}} < \lambda_{\text{visible}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}}$

Answer: B

Solution:

Wavelength of microwave is maximum then visible light then X-rays and then gamma rays so the correct order will be
$\lambda_{\text{gamma-ray}} < \lambda_{\text{Xray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

Question50

	List - I		List - II
(a)	Ultraviolet rays	(i)	Study crystal structure
(b)	Microwaves	(ii)	Greenhouse effect
(c)	Infrared rays	(iii)	Sterilizing surgical instrument
(d)	X-rays	(iv)	Radar system

Choose the correct answer from the options given below: [27-Jun-2022-Shift-1]

Options:

A. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

B. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

D. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Answer: A

Solution:

UV rays are used to sterilize surgical material.
Microwaves are used in radar system.
Infrared are used for green house effect and
X-rays are used to study crystal structure.

Question51

Given below are two statements:
Statement I : A time varying electric field is a source of changing magnetic field and vice-versa. Thus a disturbance in electric or magnetic field creates EM waves.
Statement II : In a material medium, the EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 E_0}}$. In the light of the above statements, choose the correct answer from the options given below.
[27-Jun-2022-Shift-2]

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is correct but Statement II is false

D. Statement I is incorrect but Statement II is true

Answer: C

Solution:

In a material medium speed of light is given by $v = \frac{1}{\sqrt{\varepsilon_0 \varepsilon_r \mu_0 \mu_r}}$. So statement 2 is false.

Question52

An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of $60 Vm^{-1}$. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :
[28-Jun-2022-Shift-2]

Options:

A. $E_y = 60 \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{j}\, V\ m^{-1}$

$B_z = 2 \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{k}\, T$

B. $E_y = 60 \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{j}\, V\ m^{-1}$

$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{k}\, T$

C. $E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{j}\, V\ m^{-1}$

$B_z = 60 \sin\left[\frac{\pi}{4} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{k}\, T$

D. $E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^4 (x - 4 \times 10^8 t) \right] \hat{j}\, V\ m^{-1}$

$B_z = 60 \sin\left[\frac{\pi}{4} \times 10^4 (x - 4 \times 10^8 t) \right] \hat{k}\, T$

Answer: B

Solution:

In first 3 options speed of light is $3 \times 10^8 m/\ sec$ and in the fourth option it is $4 \times 10^8 m/\ sec$.
Using
$E = CB$
We can check the option is B.

Question53

The intensity of the light from a bulb incident on a surface is $0.22 W/m^2$. The amplitude of the magnetic field in this light-wave is _____ $\times 10^{-9} T$
(Given : Permittivity of vacuum $E_0 = 8.85 \times 10^{-12} c^2 N^{-1} - m^{-2}$, speed of light in vacuum $c = 3 \times 10^8 ms^{-1}$)
[29-Jun-2022-Shift-1]

Answer: 43

Solution:

$I = \frac{1}{2}\varepsilon_0 E_0{}^2 \cdot c = \frac{1}{2}\varepsilon_0 (cB_0)^2 c$

$\Rightarrow I = \frac{1}{2}\varepsilon_0 c^3 B_0{}^2$

$\Rightarrow 0.22 = \frac{1}{2}(8.85 \times 10^{-12})(3 \times 10^8)^3 B_0{}^2$

$\Rightarrow B_0 \simeq 43 \times 10^{-9} T$

Question54

Light wave travelling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct) V m^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 ms^{-1}$)
[25-Jul-2022-Shift-2]

Options:

A. $18 \times 10^{-7} T$

B. $54 \times 10^{-7} T$

C. $54 \times 10^{-8} T$

D. $18 \times 10^{-8} T$

Answer: A

Solution:

$c = 3 \times 10^{8} m / sec$
$B = \frac{E}{c} = \frac{540}{3 \times 10^{8}} = 18 \times 10^{-7} T$

Question55

The magnetic field of a plane electromagnetic wave is given by :
$\vec{B} = 2 \times 10^{-8} \sin(0.5 \times 10^{3} x + 1.5 \times 10^{11} t) \hat{j}\ T$
The amplitude of the electric field would be :
[26-Jul-2022-Shift-1]

Options:

A. $6 Vm^{-1}$ along x-axis

B. $3 Vm^{-1}$ along z-axis

C. $6 Vm^{-1}$ along z-axis

D. $2 \times 10^{-8} Vm^{-1}$ along z-axis

Answer: C

Solution:

Speed of light $c = \frac{\omega}{k} = \frac{1.5 \times 10^{11}}{0.5 \times 10^{3}} = 3 \times 10^{8} m / sec$
So, $E_0 = B_0 c$
$= 2 \times 10^{-8} \times 3 \times 10^{8}$
$= 6 V / m$
Direction will be along z-axis.

Question56

The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000\,\pi(5x - 4 \times 10^8 t)\,T$. The amplitude of electric field will be :
[26-Jul-2022-Shift-2]

Options:

A. $15 \times 10^2 Vm^{-1}$

B. $5 \times 10^{-6} Vm^{-1}$

C. $16 \times 10^{12} Vm^{-1}$

D. $4 \times 10^2 Vm^{-1}$

Answer: D

Solution:

$B_0 = 5 \times 10^{-6}$

$v =$ Speed of wave $= \frac{4 \times 10^8}{5} = 8 \times 10^7 \left[\therefore v = \frac{W}{k} \right]$

$E_0 = vBB_0 = 40 \times 10^1$

$= 4 \times 10^2 V/m$

Question57

Identify the correct statements from the following descriptions of various properties of electromagnetic waves.
(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.
(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.
(C) Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.
(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.
(E) The ratio of amplitude of magnetic field to the amplitude of

electric field is equal to speed of light.
Choose the most appropriate answer from the options given below : [27-Jul-2022-Shift-2]

Options:

A. (D) only

B. (B) and (D) only

C. (B), (C) and (E) only

D. (A), (B) and (E) only

Answer: B

Solution:

In an EM wave:
1. $\vec{E} \perp \vec{B}$
2. $\vec{V} \equiv \vec{E} \times \vec{B}$
3. Energy is equally divided
4. $|\vec{V}| = |\vec{E}| / |\vec{B}|$

Question58

Match List - I with List - II :

List I	List II
(a)UV rays	(i)Diagnostic tool in medicine
(b)X-rays	(ii) Water purification
(c)Microwave	(iii) Communication, Radar
(d)Infrared wave	(iv) Improving visibility in foggy days

Choose the correct answer from the options given below : [29-Jul-2022-Shift-1]

Options:

A. (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

B. (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

C. (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

D. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Answer: B

Solution:

(a) uv rays - used for water purification
(b) x-rays used for diagnosing fracture
(c) Microwaves are used for mobile and radar communication
(d) Infrared waves show less scattering therefore used in foggy days
$(a-ii), (b-i), (c-iii), (d-iv)$

Question59

A radiation is emitted by 1000W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1}$ V / m. Value of x is .... (Rounded-off to the nearest integer) [Take, $\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$, $c = 3 \times 10^8 ms^{-1}$]
[26 Feb 2021 Shift 1]

Answer: 137

Solution:

Given, power of bulb, $P = 1000W$
Distance $d = 2m$
Efficiency of bulb $= 1.25\%$,
$\varepsilon_0 = 8.854 \times 10^{-12} C^2 N^{-1} m^{-1}$,
$c = 3 \times 10^8 ms^{-1}$
∴ Intensity, $(I) = \frac{\text{Power(P)}}{\text{Area(A)}} = \frac{1}{2}\varepsilon_0 E^2 c$
$\Rightarrow \quad \frac{1.25}{100} \times \frac{1000}{4\pi(2)^2} = \frac{1}{2} \times 8.854 \times 10^{-12}$

$$\Rightarrow E = \sqrt{\frac{12.5}{16\pi} \times \frac{2}{8.854 \times 10^{-4} \times 3} \times 10^{8}}$$

$= 13.7 N C^{-1} = 137 \times 10^{-1} N C^{-1}$ or $V m^{-1}$

$\therefore x = 137$

Question60

The peak electric field produced by the radiation coming from the 8W bulb at a distance of 10m is $\frac{x}{10}\sqrt{\frac{\mu_0 c}{\pi}}\frac{V}{m}$. The efficiency of the bulb is 10% and it is a point source. The value of x is .... [25 Feb 2021 Shift 2]

Answer: 2

Solution:

Given, radiation power $(P) = 8W$
Distance $(d) = 10m$

$\because$ Intensity $(I) = \frac{1}{2} c \varepsilon_0 E^2 = \frac{\text{Power (P)}}{\text{Area (A)}}$ . . . (i)

where, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} =$ speed of light in vacuum ... (ii)

and $E =$ electric field

From Eq. (ii), we get

$$\varepsilon_0 = \frac{1}{\mu_0 c^2}$$

Put this value of ε_0 in Eq. (i), we get

$$I = \frac{1}{2} C \frac{1}{\mu_0 C^2} E^2 = \frac{P}{A} \Rightarrow \frac{1}{2} \frac{1}{\mu_0 C} E^2 = \frac{P}{A}$$

$$\Rightarrow E = \sqrt{\frac{2P\mu_0 C}{A}} = \sqrt{\frac{2P\mu_0 C}{4\pi d^2}} = \sqrt{\frac{2P}{4d^2}}\sqrt{\frac{\mu_0 C}{\pi}}$$

$$= \sqrt{\frac{2 \times 8}{4 \times 100}} = \sqrt{\frac{\mu_0 C}{\pi}} = \frac{2}{10}\sqrt{\frac{\mu_0 C}{\pi}} V/m$$

$\therefore x = 2$

Question61

An electromagnetic wave of frequency 3GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be .... $\times 10^{-2}$cm. [24 Feb 2021 Shift 2]

Answer: 667

Solution:

Given, frequency of wave, $f = 3GHz = 3 \times 10^{9}Hz$
Relative permittivity, $\varepsilon_r = 2.25$
Since, $f = C / \lambda$
$\Rightarrow \lambda = \frac{C}{f} = \frac{3 \times 10^{8}}{3 \times 10^{9}} = 0.1m$
$\because \lambda_m$ (wavelength of wave in a medium) $= \lambda / \mu$ and as we know that, $\mu = \sqrt{\mu_r \varepsilon_r}$
As, dielectric is non-magnetic, $\mu_r = 1$
$\Rightarrow \mu = \sqrt{2.25} = 1.5$
$\Rightarrow \lambda_m = \frac{0.1}{1.5} = \frac{1}{15} = 0.0667m$
$= 6.67cm = 667 \times 10^{-2}cm$

Question62

List-I	List-II
A. Source of microwave frequency	1. Radioactive decay of nucleus
B. Source of infrared frequency	2. Magnetron
B. Source gamma rays	3. Inner shell electrons
C. Source of X-rays	Vibration of atoms and molecules
	5.LASER
	6. R-C circuit

choose the correct answer from the options given below [24 Feb 2021 Shift 2]

Options:

A. (A-6), (B-4), (C-1), (D-5)

B. (A-6), (B-5), (C-1), (D-4)

C. (A-2), (B-4), (C-6), (D-3)

D. (A-2), (B-4), (C-1), (D-3)

Answer: D

Solution:

As we know that;
A. Source of microwave frequency is magnetron.
B. Source of infrared frequency is vibration of atoms and molecules.
C. Source of gamma ray is radioactive decay of nucleus.
D. Source of X-ray is transition of electron in inner shells.
∴ The correct match is $A \rightarrow (2), B \rightarrow (4), C \rightarrow (1), D \rightarrow (3)$.

Question63

An electromagnetic wave of frequency 5GH z, is travelling in a medium whose relative electric permittivity and relative magnetic

permeability both are 2 . Its velocity in this medium is $\times 10^{7} \mathrm{~m} / \mathrm{s}$. [24feb2021shift1]

Answer: 15

Solution:

Given : Frequency of wave $v=5 \mathrm{GHz}=5 \times 10^{9} \mathrm{~Hz}$

Relative permittivity of the medium, $\varepsilon_{r}=2$ and relative permeability of the medium, $\mu_{r}=2$
Since speed of light in a medium is given by,
$v=\frac{1}{\sqrt{\mu \varepsilon}}=\frac{1}{\sqrt{\mu_{r} \mu_{0} \varepsilon_{r} \varepsilon_{0}}}=\frac{c}{\sqrt{\mu_{r} \varepsilon_{r}}}$
where c is speed of light is vacuum.
$\therefore v=\frac{3 \times 10^{8}}{\sqrt{4}}=\frac{30 \times 10^{7}}{2} \mathrm{~m} / \mathrm{s}=15 \times 10^{7} \mathrm{~m} / \mathrm{s}$

Question64

A plane electromagnetic wave propagating along y-direction can have the following pair of electric field (E) and magnetic field (B) components.
[18 Mar 2021 Shift 2]

Options:

A. E_{y}, B_{y} or E_{z}, B_{z}

B. E_{y}, B_{x} or E_{x}, B_{y}

C. E_{x}, B_{z} or E_{z}, B_{x}

D. E_{x}, B_{y} or E_{y}, B_{x}

Answer: C

Solution:

Electric field, magnetic field and the direction of wave propagation are mutually perpendicular to each other. For electromagnetic waves,
$E \times B = C$
The direction of wave propagation is in along +y-direction.
Therefore, the possible direction of electric field and magnetic field are (E_x, B_z) and (E_x, B_x).

Question65

A plane electromagnetic wave of frequency $100MHz$ is travelling in vacuum along the x-direction. At a particular point in space and time, $B = 2.0 \times 10^{-8}\hat{k}T$ (where, $\hat{k}$ is unit vector along z-direction). What is E at this point? [18 Mar 2021 Shift 1]

Options:

A. $0.6\hat{j}V/m$

B. $6.0\hat{k}V/m$

C. $6.0\hat{j}V/m$

D. $0.6\hat{k}V/m$

Answer: C

Solution:

Given, $f = 100MHz$
$B = (2.0 \times 10^{-8})\hat{k}T$
The speed of the light in free space,
$c = (3 \times 10^{s})\hat{i}m/s$
(the direction of wave propagates in x-direction)
Magnitude of electric field, $E = Bc = 2 \times 10^{-8} \times 3 \times 10^{8}$
$\Rightarrow E = 6V/m$
If $\hat{n}$ be the direction of E, then $\hat{i} = \hat{n} \times \hat{k}$
$\Rightarrow \hat{n} = \hat{j}$
Hence, the electric field at this point is $6\hat{j}V/m$.

Question66

The electric field intensity produced by the radiation coming from a 100W bulb at a distance of 3m is E. The electric field intensity produced by the radiation coming from 60W at the same distance is $\sqrt{\frac{x}{5}}E$, where the value of x is ....

[17 Mar 2021 Shift 2]

Answer: 3

Solution:

Intensity of the electromagnetic radiation is given as

$$I = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{\text{Power}}{\text{Area}} = U_{avg}C = \frac{1}{2}C\varepsilon_0 E_0^2$$

Here,

U_{avg} = average energy density,

C = speed of radiation in air (or vacuum),

ε_0 = permittivity of the free space

and E_0 = peak value of the electric field.

$$\Rightarrow P \propto E_0^2 \Rightarrow \frac{P_1}{P_2} = \frac{E_1^2}{E_2^2}$$

$$\Rightarrow \frac{E_1}{E_2} = \sqrt{\frac{P_1}{P_2}}$$

Substituting the values in the above equation, we get

$$\Rightarrow \frac{E_1}{E_2} = \sqrt{\frac{60}{100}}$$

$$\Rightarrow E_1 = \sqrt{\frac{3}{5}}E_2 = \sqrt{\frac{3}{5}} = E \quad (\because E_2 = E)$$

Comparing with $\sqrt{\frac{x}{5}}E$, we get

$x = 3$

Question67

Seawater at a frequency $f = 9 \times 10^2 Hz$, has permittivity $\varepsilon = 80\varepsilon_0$ and resistivity $r = 0.25\Omega - m$. Imagine a parallel plate capacitor is immersed in seawater and is driven by an alternating voltage source $V(t) = V_0 \sin(2\pi f t)$. Then, the conduction current density becomes 10^x times the displacement current density after time $t = \frac{1}{800}$s. The value of x is ....

$\left(\text{Take }, \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 N - m^2 C^{-2} \right)$

[17 Mar 2021 Shift 2]

Answer: 6

Solution:

As we know, conduction current,

$$I_c = \frac{[v_0 \sin(2\pi f t)]}{\rho \frac{d}{A}}$$

$$\Rightarrow I_c = \frac{A[v_0 \sin(2\pi f t)]}{\rho d} ..(i)$$

As we know the displacement current equation,

$$I_d = \frac{\varepsilon_0 \varepsilon_t A}{d} \frac{d}{dt}[v_0 \sin(2\pi f t)]$$

$$I_d = \frac{\varepsilon_0 \varepsilon_t A}{d} v_0 (2\pi f)[\cos(2\pi f t)] ...(ii)$$

Divide Eq. (ii) by Eq. (i), we get

$$\frac{I_d}{I_c} = \varepsilon_0 \varepsilon_r \times \rho 2\pi f \cot(2\pi f t)$$

Substituting the values in the above equation, we get

$$\frac{I_d}{I_c} = \frac{1}{4\pi \times 9 \times 10^9} \times (80) \times 2\pi \times 900(0.25) \cot\left(2\pi(900) \frac{1}{800} \right)$$

$$\frac{I_d}{I_c} = \frac{1}{10^6}$$

Comparing with 10^x, we get $x = 6$.

Question68

A plane electromagnetic wave of frequency 500M H z is travelling in vacuum along y-direction. At a particular point in space and time, $B = 8.0 \times 10^{-8} \hat{z} T$. The value of electric field at this point is (speed of light $= 3 \times 10^{8} ms^{-1}$; $\hat{x}$, $\hat{y}$, $\hat{z}$ are unit vectors along x, y and z-direction). [16 Mar 2021 Shift 1]

Options:

A. $-24\hat{x} V/m$

B. $2.6\hat{x} V/m$

C. $24\hat{x} V/m$

D. $-2.6\hat{x} V/m$

Answer: A

Solution:

Given, frequency, $f = 500 MHz = 5 \times 10^{8} Hz$
$B = 8.0 \times 10^{-8} \hat{Z} T$
∴ Magnitude of peak value of magnetic field is given by
$B_0 = 8 \times 10^{-8} T$
We know that, $\frac{E_0}{B_0} = C$
where, E_0 is the magnitude of peak value of electric field and c is the speed of electromagnetic wave in air (or vacuum).
$\Rightarrow E_0 = cB_0$
$= 3 \times 10^{8} \times 8 \times 10^{-8} = 24 V/m$
Since, the direction of propagation of electromagnetic wave is perpendicular to the direction of E and B both.
∴ Direction of propagation is given by $\hat{E} \times \hat{B}$.
As, the wave is travelling in y-direction,
and the magnetic field is in z-direction.
$\Rightarrow \hat{E} \times \hat{z} = \hat{y}$
$\hat{E} = -\hat{x}$
∴ The value of electric field will be $-24\hat{x} V/m$.

Question69

For an electromagnetic wave travelling in free space, the relation between average energy densities due to electric (U_e) and magnetic (U_m) fields is
[16 Mar 2021 Shift 1]

Options:

A. $U_e = U_m$

B. $U_e > U_m$

C. $U_e < U_m$

D. $U_e \neq U_m$

Answer: A

Solution:

We know that,
$U_e = \frac{1}{2}\varepsilon_0 E^2$....(1)
where, $U_e =$ average energy density due to electric field,
$\varepsilon_0 =$ electrical permittivity of free space
and $E =$ electric field.
and $U_m = \frac{B^2}{2\mu_0}$.....(2)
where, $U_m =$ average energy density due to magnetic field, $\mu_0 =$ magnetic permeability of free space
and $B =$ magnetic field.
Also, $E = CB \quad \left(\because C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}\right)$
$\Rightarrow E = \frac{B}{\sqrt{\mu_0 \varepsilon_0}}$
Put this value in Eq. (1), we get
$U_e = \frac{1}{2}\varepsilon_0 \times \frac{1}{\mu_0 \varepsilon_0} B^2$
$\Rightarrow U_e = \frac{B^2}{2\mu_0}$
$\Rightarrow U_e = U_m$ [using Eq. (2)]

Question70

The relative permittivity of distilled water is 81 . The velocity of light in it will be :
(Given $\mu_r = 1$)
[27 Jul 2021 Shift 1]

Options:

A. $4.33 \times 10^7 \mathrm{m/s}$

B. $2.33 \times 10^7 \mathrm{m/s}$

C. $3.33 \times 10^7 \mathrm{m/s}$

D. $5.33 \times 10^7 \mathrm{m/s}$

Answer: C

Solution:

$$V = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$
$$= 3.33 \times 10^7 \mathrm{m/s}$$

Question71

Intensity of sunlight is observed as $0.092 \mathrm{W\,m^{-2}}$ at a point in free space. What will be the peak value of magnetic field at that point?
($\varepsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2 N^{-1} m^{-2}}$)
[22 Jul 2021 Shift 2]

Options:

A. $2.77 \times 10^{-8} \mathrm{T}$

B. $1.96 \times 10^{-8} \mathrm{T}$

C. $8.31 \mathrm{T}$

D. $5.88 \mathrm{T}$

Answer: A

Solution:

$$I_{avg} = \frac{B_0^{\ 2}C}{2\mu_0} \ \& \ \frac{1}{\mu_0} = \epsilon_0 C^2$$

$$I = \frac{B_0^{\ 2}}{2}\epsilon_0 C^3$$

$$B_0 = \sqrt{\frac{2I}{\epsilon_0 C^3}}$$

$$B_0 = 2.77 \times 10^{-8} T$$

Question72

In an electromagnetic wave the electric field vector and magnetic field vector are given as $\vec{E} = E_0 \hat{i}$ and $\vec{B} = B_0 \hat{k}$ respectively. The direction of propagation of electromagnetic wave is along:
[20 Jul 2021 Shift 2]

Options:

A. $(\hat{k})$

B. $\hat{J}$

C. $(-\hat{k})$

D. $(-\hat{j})$

Answer: D

Solution:

Direction of propagation $= \vec{E} \times \vec{B} = \hat{i} \times \hat{k} = -\hat{j}$

Question73

A light beam is described by

$E = 800 \sin\left(\omega t - \frac{x}{c}\right)$. An electron is allowed to move normal to the propagation of light beam with a speed $3 \times 10^7 ms^{-1}$. What is the maximum magnetic force exerted on the electron?
[26 Aug 2021 Shift 2]

Options:

A. $1.28 \times 10^{-17} N$

B. $1.28 \times 10^{-18} N$

C. $12.8 \times 10^{-17} N$

D. $12.8 \times 10^{-18} N$

Answer: 0

Solution:

Equation for light beam is given as
$E = 800 \sin\left(\omega t - \frac{x}{C}\right)$
Here, c is speed of light.
Comparing with standard equation, we get
$E_0 = 800 Vm^{-1}$
As we know, the magnitude of magnetic field is given as
$E_0 = B_0 C$
$\Rightarrow 800 = B_0 \times 3 \times 10^8$
$\Rightarrow B_0 = \frac{800}{3 \times 10^8} T$
Maximum magnetic force exerted on electron will be
$F_{max} = ev B_0$
$= (1.6 \times 10^{-19}) \times (3 \times 10^7) \times \frac{800}{3 \times 10^8}$ (Given, $v = 3 \times 10^7 ms^{-1}$)
$F_{max} = 12.8 \times 10^{-18}$ N or 1.28×10^{-17}

Question74

The magnetic field vector of an electromagnetic wave is given by $B = B_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz - \omega t)$ T where $\hat{i}$, $\hat{j}$ represents unit vector along X and Y-axis respectively. At t = 0, two electric charges q_1 of 4πC and q_2 of 2π C located at $\left(0, 0, \frac{\pi}{k}\right)$ and $\left(0, 0, \frac{3\pi}{k}\right)$ respectively, have the same velocity of $0.5c\hat{i}$.
(where, c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is
[31 Aug 2021 Shift 2]

Options:

A. $2\sqrt{2} : 1$

B. $1 : \sqrt{2}$

C. $2 : 1$

D. $\sqrt{2} : 1$

Answer: C

Solution:

The equation of magnetic field vector of an electromagnetic field is
$$B = \frac{B_0}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)\cos(kz - \omega t)\ T$$
Electric charges
q_1 at $P\left(0, 0, \frac{\pi}{k}\right) = 4\pi C$

q_2 at $Q\left(0, 0, \frac{3\pi}{k}\right) = 2\pi C$

At $t = 0$ and $P\left(0, 0, \frac{\pi}{k}\right)$, where $z = \frac{\pi}{k}$
Magnetic field vector,
$$B = \frac{B_0}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)\cos\left(k\frac{\pi}{k} - 0\right)\ T$$
$$= \frac{B_0}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)\cos\pi\ T = -\frac{B_0}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)T$$
and at $t = 0$ and $Q\left(0, 0, \frac{3\pi}{k}\right)$, where $z = \frac{3\pi}{k}$. Magnetic field vector,
$$B = \frac{B_0}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)\cos\left(k\frac{3\pi}{k} - 0\right)$$

$= -\frac{B_0}{\sqrt{2}}(\hat{i}+\hat{j})T$ [∵ Using cos π = cos 3π = − 1]

Velocity of charges q_1 and q_2, $v = 0.5c\hat{i}$

We know that, force on charge in magnetic field is given by $F = q(v \times B)$

∴ For q_1, $F_1 = 4\pi\left[0.5c\hat{i} \times \left\{-\frac{B_0}{\sqrt{2}}(\hat{i}+\hat{j})\right\}\right]N$

$F_1 = -\frac{2\pi c B_0}{\sqrt{2}}\hat{k}N$

Similarly, for q_2

$\Rightarrow F_2 = 2\pi\left[0.5c\hat{i} \times \left\{-\frac{B_0}{\sqrt{2}}(\hat{i}+\hat{j})\right\}\right]N$

$\Rightarrow F_2 = -\frac{\pi c B_0}{\sqrt{2}}\hat{k}N$

Hence, ratio of magnitudes of F_1 to F_2

$$\frac{F_1}{F_2} = \frac{\frac{2\pi c B_0}{\sqrt{2}}}{\frac{\pi c B_0}{\sqrt{2}}} = \frac{2}{1}$$

$\Rightarrow F_1 : F_2 = 2 : 1$

Question75

The electric field in an electromagnetic wave is given by $E = (50NC^{-1})\sin\omega\left(t-\frac{x}{c}\right)$.The energy contained in a cylinder of volume V is 5.5×10^{-12}J. The value of V is .... cm^3. (Given $\varepsilon_0 = 8.85\times10^{-12}C^2N^{-1}m^{-2}$). [31 Aug 2021 Shift 1]

Answer: 500

Solution:

Given, electric field in electromagnetic wave,

$E = 50\sin\omega\left(t-\frac{x}{c}\right)$

Energy contained in cylinder,

$U = 5.5 \times 10^{-12} J$
Let, volume = V
and permittivity in free space,
$\varepsilon_0 = 8.85 \times 10^{-12} N^{-1} C^2 m^{-2}$
As we know that,
$\frac{\text{Energy}(U)}{\text{Volume}(V)} = \frac{1}{2}\varepsilon_0 E_0{}^2$
$\therefore V = \frac{2U}{\varepsilon_0 E_0{}^2} = \frac{2 \times 5.5 \times 10^{-12}}{8.854 \times 10^{-12} \times (50)^2}$
$= \frac{11}{8.854 \times 50^2} = 4.97 \times 10^{-4}$
$= 497 \times 10^{-6} = 500 cm^3$

Question76

A plane electromagnetic wave with frequency of 30 MHz travels in free space. At particular point in space and time, electric field is 6V/m. The magnetic field at this point will be $x \times 10^{-8}$T. The value of x is.
[27 Aug 2021 Shift 2]

Answer: 2

Solution:

Given, frequency of electromagnetic wave (f)
$= 30 MHz$
$= 30 \times 10^6 Hz$
Electric field, $E = 6V/m$
Magnetic field, $B = x \times 10^{-8} T$
Speed of light in air, $c = 3 \times 10^8 ms^{-1}$
We know that, $B = \frac{E}{C}$
$\Rightarrow B = \frac{6}{3 \times 10^8}$
$\Rightarrow B = 2 \times 10^{-8} T$
$\therefore x = 2$

Question77

Electric field in a plane electromagnetic wave is given by
$E = 50 \sin(500x - 10 \times 10^{10} t)\ V/m$.
The velocity of electromagnetic wave in this medium is
(Given, c = speed of light in vacuum)
[27 Aug 2021 Shift 1]

Options:

A. $\frac{3}{2}c$

B. c

C. $\frac{2}{3}c$

D. $\frac{c}{2}$

Answer: C

Solution:

Given, equation of electromagnetic wave,
$E = 50 \sin(500x - 10 \times 10^{10} t)\ V/m$
The standard equation of electromagnetic wave,
$E = E_0 \sin(kx - \omega t)\ V/m$
Comparing with standard equation, we get
$E_0 = 50 V/m$, $\omega = 10^{10}\ rad\ s^{-1}$, $k = 500\ rad\ m^{-1}$
Speed of wave in medium will be

$$v = \frac{\omega}{k}$$
$$= \frac{10 \times 10^{10}}{500}$$
$$= 2 \times 10^8 ms^{-1}$$
$$= \frac{2}{3} \times 3 \times 10^8 m/s$$
$$= \frac{2}{3}c\ [\because c = 3 \times 10^8 m/s = \text{speed of light}]$$

Question78

The electric field in a plane electromagnetic wave is given by
E = 200cos

$$\left[\left(\frac{0.5\times10^{3}}{m}\right)x-\left(1.5\times10^{11}\ \frac{rad}{s}\times t\right)\right]\ \frac{v}{m}\hat{j}$$

If this wave falls normally on a perfectly reflecting surface having an area of 100cm^2. If the radiation pressure exerted by the EM wave on the surface during a 10 min exposure is $\frac{x}{10^9}$. Find the value of x.

[26 Aug 2021 Shift 1]

Answer: 354

Solution:

The electric field in a plane electromagnetic wave is given by

$$E=200\cos\left[\frac{(0.5\times10^{3})x}{m}-\left(1.5\times10^{11}\ \frac{rad}{s}\times t\right)\right]\frac{V}{m}\hat{j}$$

$\therefore E_0=200V/m$

Intensity provided by electric field is given by

$$I=\frac{1}{2}\varepsilon_0E_0{}^2C$$

The radiation pressure exerted by electromagnetic wave on surfaceis given by

$$P=\frac{2I}{c}=\frac{2\times\frac{1}{2}\varepsilon_0E_0{}^2C}{c}=\varepsilon_0E_0{}^2$$

$$P=\varepsilon_0E_0{}^2$$

$$=8.85\times10^{-12}\times(200)^2$$

$$=\frac{354}{10^9}$$

According to question,

$$\frac{x}{10^9}=\frac{354}{10^9}$$

$\therefore$ x = 354

Question79

Electric field of plane electromagnetic wave propagating through a non-magnetic medium is given by $E=20\cos(2\times10^{10}t-200x)\ V/m$.

The dielectric constant of the medium is equal to (Take, $\mu_r = 1$) [1 Sep 2021 Shift 2]

Options:

A. 9

B. 2

C. $\frac{1}{3}$

D. 3

Answer: A

Solution:

Given, electric field,
$E = 20\cos(2 \times 10^{10}t - 200x)\ V/m$
Comparing with the standard equation,
$E = E_0\cos(\omega t - kx)\ V/m$, we get
Wave constant, k = 200
Angular frequency, $\omega = 2 \times 10^{10}\ rad/s$
Speed of the wave, $v = \frac{\omega}{k} = \frac{2 \times 10^{10}}{200} = 10^8 m/s$
Refractive index, $\mu = \frac{C}{V} = \frac{3 \times 10^8}{10^8} = 3$
As we know the relation between the refractive index and dielectricconstant,
$\mu = \sqrt{\varepsilon_r \mu_r}$
Substituting the value in the above equations, we get
$3 = \sqrt{\varepsilon_r (1)}$
$\varepsilon_r = 9$
Thus, the dielectric constant of the medium is 9.

Question80

The electric fields of two plane electromagnetic plane waves in vacuum are given by
$\vec{E}_1 = E_0 \hat{j}\cos(\omega t - kx)$ and $\vec{E}_2 = E_0 \hat{k}\cos(\omega t - ky)$.
At t = 0, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced

by the particle is:
[9 Jan 2020, I]

Options:

A. $E_0 q(0.8\hat{i} - \hat{j} + 0.4\hat{k})$

B. $E_0 q(0.4\hat{i} - 3\hat{j} + 0.8\hat{k})$

C. $E_0 q(-0.8\hat{i} + \hat{j} + \hat{k})$

D. $E_0 q(0.8\hat{i} + \hat{j} + 0.2\hat{k})$

Answer: D

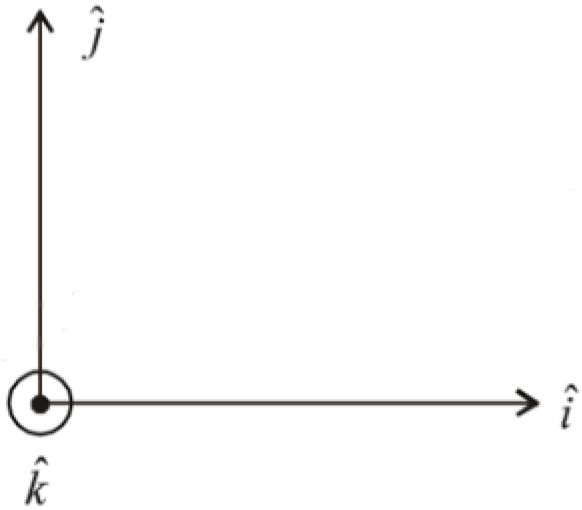
Solution:

Given: $\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx)$

i.e., Travelling in $+$ ve x -direction $\vec{E} \times \vec{B}$ should be in x direction

$\therefore \vec{B}$ is in $\hat{K}$

$\therefore \vec{B}_1 = \frac{E_0}{C} \cos(\omega t - kx) \hat{k} \left(\because B_0 = \frac{E_0}{C} \right)$



$\vec{E}_2 = E_0 \hat{k} \cos(\omega t - ky)$

$\vec{B}_2 = \frac{E_0}{C} \hat{i} \cos(\omega t - ky)$

$\therefore$ Travelling in $+$ ve y -axis

$\vec{E} \times \vec{B}$ should be in y -axis

$\therefore$ Net force $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$

$q\left(\vec{E}_1+\vec{E}_2\right)+q\left(0.8c\hat{j}\times\left(\hat{B}_1+\hat{B}_2\right)\right)$

If $t=0$ and $x=y=0$

$\vec{E}_1=E_0\hat{j},\quad \vec{E}_2=E_0\hat{k}$

$\vec{B}_1=\frac{E_0}{c}\hat{k},\quad \vec{B}_2=\frac{E_0}{c}\hat{i}$

$\therefore \vec{F}_{net}=qE_0\left(\hat{j}+\hat{k}\right)+q\times0.8c\times\frac{E_0}{C}\hat{j}\times\left(\hat{k}+\hat{i}\right)$

$=qE_0\left(\hat{j}+\hat{k}\right)+0.8qE_0\left(\hat{i}-\hat{k}\right)$

$=qE_0\left(0.8\hat{i}+\hat{j}+0.2\hat{k}\right)$

Question81

A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant):
[9 Jan 2020, II]

Options:

A. $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

B. $B_0\frac{\hat{j}-\hat{i}}{\sqrt{2}}\cos\left(\omega t+k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

C. $B_0\hat{k}\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

D. $B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

Answer: A

Solution:

Direction of polarisation $= \hat{E} = \hat{k}$

Direction of propagation $= \hat{E} \times \hat{B} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

But $\vec{E} \cdot \vec{B} = 0 \therefore \hat{B} = \frac{\hat{i} - j}{\sqrt{2}}$

Question82

A plane electromagnetic wave of frequency 25GH z is propagating in vacuum along the z -direction. At a particular point in space and time, the magnetic field is given by $\vec{B} = 5 \times 10^{-8}\hat{j}T$. The corresponding electric field $\hat{E}$ is (speed of light $c = 3 \times 10^{8}ms^{-1}$) [8 Jan 2020, II]

Options:

A. $1.66 \times 10^{-16}\hat{i}V / m$

B. $-1.66 \times 10^{-16}\hat{i}V / m$

C. $-15\hat{i}V / m$

D. $15\hat{i}V / m$

Answer: D

Solution:

Amplitude of electric field (E) and Magnetic field (B) of an electromagnetic wave are related by the relation

$\frac{E}{B} = c$

$\Rightarrow E = Bc$

$\Rightarrow E = 5 \times 10^{-8} \times 3 \times 10^{8} = 15N / C$

$\Rightarrow \vec{E} = 15\hat{i}V / m$

Question83

If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{j}$ T, then what will be expression for electric field?
[7 Jan 2020, I]

Options:

A. $\vec{E} = \left(60 \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{k}\, v/m\right)$

B. $\vec{E} = \left(9 \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{k}\, v/m\right)$

C. $\vec{E} = \left(3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{k}\, v/m\right)$

D. $\vec{E} = \left(-3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{k}\, v/m\right)$

Answer: B

Solution:

Given, $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\hat{j}\, t$
Using, $E_0 = B_0 \times C = 3 \times 10^{-8} \times 3 \times 10^{8} = 9V/m$
$\therefore$ Electric field $\vec{E} = 9 \sin(1.6 \times 10^{3}x + 48 \times 10^{10}t)\vec{k}\ V/m$

Question84

The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$
At $t = 0$, a positively charged particle is at the point $(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at $(t = 0)$ is $v_0\hat{k}$, the force acting on it due to the wave is:
[7 Jan 2020, II]

Options:

A. parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

B. zero

C. antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

D. parallel to $\hat{k}$

Answer: C

Solution:

$$\text{At } t=0,\ z=\frac{\pi}{k}$$
$$\therefore \vec{E}=\frac{E_0}{\sqrt{2}}(\hat{i}+\hat{j})\cos[\pi]=-\frac{E_0}{\sqrt{2}}(\hat{i}+\hat{j})$$
$$\hat{F}_E=q\hat{E}$$

Force due to electric field will be in the direction $\frac{-(\hat{i}+\hat{j})}{\sqrt{2}}$

Force due to magnetic field is in direction

$q(\vec{v}\times\vec{B})$ and $\vec{v}\parallel\vec{k}$. Therefore, it is parallel to $\vec{E}$.

$\Rightarrow \vec{F}_{net}=\vec{F}_E+\vec{F}_B$ is antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Question85

For a plane electromagnetic wave, the magnetic field at a point x and time t is

$\vec{B}(x,t)=[1.2\times10^{-7}\sin(0.5\times10^{3}x+1.5\times10^{11}t)\hat{k}]$ T

The instantaneous electric field $\vec{E}$ corresponding to B is: (speed of light $c=3\times10^{8}ms^{-1}$)

[Sep. 06, 2020 (II)]

Options:

A. $\vec{E}(x,t)=[-36\sin(0.5\times10^{3}x+1.5\times10^{11}t)\hat{j}]\ \frac{V}{m}$

B. $\vec{E}(x,t)=[36\sin(1\times10^{3}x+0.5\times10^{11}t)\hat{j}]\ \frac{V}{m}$

C. $\vec{E}(x, t) = [36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{k}] \frac{V}{m}$

D. $\vec{E}(x, t) = [36 \sin(1 \times 10^3 x + 0.5 \times 10^{11} t)\hat{i}] \frac{V}{m}$

Answer: A

Solution:

Relation between electric field E_0 and magnetic field B_0 of an electromagnetic wave is given by

$c = \frac{E_0}{B_0}$ (Here ,c = Speed of light)

$\Rightarrow E_0 = B_0 \times c = 1.2 \times 10^{-7} \times 3 \times 10^8 = 36$

As the wave is propagating along x -direction, magnetic field is along z -direction

and $(\hat{E} \times \hat{B}) \parallel \hat{C}$

$\therefore E$ should be along y -direction.

So, electric field $\vec{E} = E_0 \sin \vec{E} . (x, t)$

$= [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{j}] \frac{V}{m}$

Question86

An electron is constrained to move along the y -axis with a speed of 0.1c(c is the speed of light) in the presence of electromagnetic wave, whose electric field is $\vec{E} = 30\hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x)$ V / m. The maximum magnetic force experienced by the electron will be : (given $c = 3 \times 10^8 ms^{-1}$ & electron charge $= 1.6 \times 10^{-19} C$) [Sep. 05,2020 (I)]

Options:

A. $3.2 \times 10^{-18} N$

B. $2.4 \times 10^{-18} N$

C. $4.8 \times 10^{-19} N$

D. $1.6 \times 10^{-19} N$

Answer: C

Solution:

In electromagnetic wave, $\frac{E_0}{B_0} = C$

$\therefore$ Maximum value of magnetic field, $B_0 = \frac{E_0}{C}$

$F_{max} = qV B_{max} \sin 90° = \frac{qV_0 E_0}{C}$

(Given $V_0 = 0.1C$ and $E_0 = 30$)

$= \frac{1.6 \times 10^{-19} \times 0.1 \times 3 \times 10^8 \times 30}{3 \times 10^8} = 4.8 \times 10^{-19} N$

Question87

The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$

Its magnetic field will be given by:
[Sep. 04,2020 (II)]

Options:

A. $\frac{E_0}{c}(-\hat{x} + \hat{y}) \sin(kz - \omega t)$

B. $\frac{E_0}{c}(\hat{x} + \hat{y}) \sin(kz - \omega t)$

C. $\frac{E_0}{c}(\hat{x} - \hat{y}) \sin(kz - \omega t)$

D. $\frac{E_0}{c}(\hat{x} - \hat{y}) \cos(kz - \omega t)$

Answer: A

Solution:

$\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$

Direction of propagation of em wave $= +\hat{k}$

Unit vector in the direction of electric field, $\hat{E} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

The direction of electromagnetic wave is perpendicular to both electric and magnetic field.

$\therefore \hat{k} = \hat{E} \times \hat{B}$

$\Rightarrow \hat{k} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) \times \hat{B} \Rightarrow \hat{B} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}}$

$\therefore \vec{B} = \frac{E_0}{c}(-\hat{x} + \hat{y}) \sin(kz - \omega t)$

Question88

The magnetic field of a plane electromagnetic wave is $\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{i}$ T where $c = 3 \times 10^{8} ms^{-1}$ is the speed of light. The corresponding electric field is : [Sep. 03, 2020 (I)]

Options:

A. $\vec{E} = 9 \sin[200\pi(y + ct)] \hat{k} V/m$

B. $\vec{E} = -10^{-6} \sin[200\pi(y + ct)] \hat{k} V/m$

C. $\vec{E} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{k} V/m$

D. $\vec{E} = -9 \sin[200\pi(y + ct)] \hat{k} V/m$

Answer: D

Solution:

(d) Given: $\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{i}$ T

$\therefore B_0 = 3 \times 10^{-8}$

$E_0 = CB_0 \Rightarrow E_0 = 3 \times 10^{8} \times 3 \times 10^{-8} = 9V/m$

Directiono fwave propagation

$(\vec{E} \times \vec{B}) \parallel \vec{C}$

$\hat{B} = \hat{i}$ and $\hat{C} = -\hat{j} \therefore \hat{E} = -\hat{k}$

$\therefore \vec{E} = E_0 \sin[200\pi(y + ct)](-\hat{k}) V/m$

or, $\vec{E} = -9 \sin[200\pi(y + ct)] \hat{k} V/m$

Question89

The electric field of a plane electromagnetic wave propagating along the x direction in vacuum is $\vec{E} = E_0 \hat{j} \cos(\omega t - kx)$. The magnetic field $\vec{B}$, at the moment t = 0 is:
[Sep. 03, 2020 (II)]

Options:

A. $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \varepsilon_0}} \cos(kx) \hat{k}$

B. $\vec{B} = E_0 \sqrt{\mu_0 \varepsilon_0} \cos(kx) \hat{j}$

C. $\vec{B} = E_0 \sqrt{\mu_0 \varepsilon_0} \cos(kx) \hat{k}$

D. $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \varepsilon_0}} \cos(kx) \hat{j}$

Answer: C

Solution:

Relation between electric field and magnetic field for an electromagnetic wave in vacuum is $B_0 = \frac{E_0}{c}$.

In free space, its speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

Here, $\mu_0 =$ absolute permeability, $\varepsilon_0 =$ absolute permittivity

$\therefore B_0 = \frac{E_0}{c} = \frac{E_0}{1/\sqrt{\mu_0 \varepsilon_0}} = E_0 \sqrt{\mu_0 \varepsilon_0}$

As the electromagnetic wave is propagating along x direction and electric field is along y direction.

$\therefore \hat{E} \times \hat{B} \parallel \hat{C}$ (Here, $\hat{C} =$ direction of propagation of wave)

$\therefore \vec{B}$ should be in $\hat{k}$ direction.

$\therefore B = E_0 \sqrt{\mu_0 \varepsilon_0} \cos(\omega t - kx) \hat{k}$

At $t = 0$

$B = E_0 \sqrt{\mu_0 \varepsilon_0} \cos(kx) \hat{k}$

Question90

A plane electromagnetic wave, has frequency of 2.0×10^{10} Hz and its energy density is $1.02 \times 10^{-8} J/m^3$ in vacuum. The amplitude of the magnetic field of the wave is close to $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \frac{N m^2}{C^2}\right.$ and speed of light $= 3 \times 10^8 ms^{-1}$)
[Sep. 02, 2020 (I)]

Options:

A. 150 nT

B. 160 nT

C. 180 nT

D. 190 nT

Answer: B

Solution:

Energy density $= \frac{1}{2}\frac{B^2}{\mu_0}$

$\Rightarrow B = \sqrt{2 \times \mu_0 \times \text{ Energy density}}$

$mu_0 = \frac{1}{C^2\varepsilon_0} = 4\pi \times 10^{-7}$

$\therefore B = \sqrt{2 \times 4\pi \times 10^{-7} \times 1.02 \times 10^{-8}} = 160 \times 10^{-9}$

$= 160nT$

Question91

In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$ respectively. What is the unit vector along direction of propagation of the wave.
[Sep. 02, 2020 (II)]

Options:

A. $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$

B. $\frac{1}{\sqrt{2}}\left(\hat{j}+\hat{k}\right)$

C. $\frac{1}{\sqrt{5}}\left(\hat{i}+2\hat{j}\right)$

D. $\frac{1}{\sqrt{5}}\left(2\hat{i}+\hat{j}\right)$

Answer: A

Solution:

Electromagnetic wave will propagate perpendicular to the direction of Electric and Magnetic fields
$\hat{C}=\hat{E}\times\hat{B}$
Here unit vector $\hat{C}$ is perpendicular to both $\hat{E}$ and $\hat{B}$ Given, $\vec{E}=\hat{k}, \vec{B}=2\hat{i}-2\hat{j}$

$$\hat{C}=\hat{E}\times\hat{B}=\frac{1}{\sqrt{2}}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 0 & 1\\ 1 & -1 & 0\end{vmatrix}=\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

$$\Rightarrow\hat{C}=\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

Question92

TOPIC 2 - Electromagnetic Spectrum
The correct match between the entries in column I and column II

are :

I	II
Radiation	Wavelength
(A) Microwave	(i) 100 m
(B) Gamma rays	(ii) 10^{-15} m
(C) A.M. radio waves	(iii) 10^{-10} m
(D) X-rays	(iv) 10^{-3} m

[Sep. 05, 2020 (II)]

Options:

A. (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

B. (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

C. (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv)

D. (A)-(iv), (B)-(ii), (C)-(i), (D)-(iii)

Answer: D

Solution:

Energy sequence of radiations is
$E_{\gamma-\text{ Rays}} > E_{\text{X-Rays}} > E_{\text{microwave}} > E_{\text{AM Radiowaves}}$
$\therefore \lambda_{\gamma-\text{Rays}} < \lambda_{\text{X -Rays}} < \lambda_{\text{microwave}} < \lambda_{\text{AM Radiowaves}}$
From the above sequence, we have
(a) Microwave $\rightarrow 10^{-3}\text{m}$ (iv)
(b) Gamma Rays $\rightarrow 10^{-15}\text{m}$ (ii)
(c) AM Radio wave $\rightarrow 100\text{m}$ (i)
(d) X-Rays $\rightarrow 10^{-10}\text{m}$ (iii)

Question93

Chosse the correct option relating wavelengths of different parts of electromagnetic wave spectrum :

[Sep. 04, 2020 (I)]

Options:

A. $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{X -rays}}$

B. $\lambda_{\text{radio waves}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{X -rays}}$

C. $\lambda_{\text{X -rays}} < \lambda_{\text{micro waves}} > \lambda_{\text{radio waves}} < \lambda_{\text{visible}}$

D. $\lambda_{\text{visible}} > \lambda_{\text{X -rays}} > \lambda_{\text{radio waves}} > \lambda_{\text{micro waves}}$

Answer: B

Solution:

The orderly arrangement of different parts of EM wave in decreasing order of wavelength is as follows:
$\lambda_{\text{radio waves}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{X -rays}}$

Question94

The mean intensity of radiation on the surface of the Sun is about $10^8 W/m^2$. The rms value of the corresponding magnetic field is closest to :
[12 Jan 2019, II]

Options:

A. $1T$

B. $10^2 T$

C. $10^{-2} T$

D. $10^{-4} T$

Answer: D

Solution:

$I = \frac{B_0^2}{2\mu_0} . C$

$\Rightarrow \frac{B_0^2}{2} = \frac{I\,mu_0}{C}$

$\Rightarrow B_{rms} = \sqrt{\frac{I\,\mu_0}{C}}$

$= \sqrt{\frac{10^8 \times 4\pi \times 10^{-7}}{3 \times 10^8}}$

$\simeq 6 \times 10^{-4} T$

Which is closest to 10^{-4}

Question95

An electromagnetic wave of intensity $50 W\ m^{-2}$ enters in a medium of refractive index 'n' without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by:
[11 Jan 2019, I]

Options:

A. $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$

B. $(\sqrt{n}, \sqrt{n})$

C. $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$

D. $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$

Answer: C

Solution:

The speed of electromagnetic wave in free space is given by

$C = \frac{1}{\sqrt{\mu_0 \in_0}}$ .....(i)

In medium, $v = \frac{1}{\sqrt{k \in_0 \mu_0}}$ ......(ii)
Dividing equation (i) by (ii), we get
$\therefore \frac{C}{V} = \sqrt{k} = n$
$\frac{1}{2} \in_0 E_0^2 C =$ intensity $= \frac{1}{2} \in_0 kE^2 v$
$\therefore E_0^2 C = kE^2 v$
$\Rightarrow \frac{E_0^2}{E^2} = \frac{kV}{C} = \frac{n^2}{n} \Rightarrow \frac{E_0}{E} = \sqrt{n}$
similarly
$\frac{B_0^2 C}{2\mu_0} = \frac{B^2 v}{2\mu_0} \Rightarrow \frac{B_0}{B} = \frac{1}{\sqrt{n}}$

Question96

A 27mW laser beam has a cross-sectional area of $10 mm^2$ The magnitude of the maximum electric field in this electromagnetic wave is given by :
[Given permittivity of space $\in_0 = 9 \times 10^{-12}$ SI units, Speed of light $c = 3 \times 10^8 m/s$]
[11 Jan 2019, II]

Options:

A. 2 kV/m

B. 0.7 kV/m

C. 1 kV/m

D. 1.4 kV/m

Answer: D

Solution:

EM wave intensity
$\Rightarrow I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2} \in_0 E_0^2 c$
[where $E_0 =$ maximum electric field]

$$\Rightarrow \frac{27\times10^{-3}}{10\times10^{-6}} = \frac{1}{2}\times9\times10^{-12}\times E_0^{\;2}\times3\times10^{8}$$
$$\Rightarrow E_0 = \sqrt{2}\times10^{3}\mathrm{kV/m} = 1.4\mathrm{kV/m}$$

Question97

If the magnetic field of a plane electromagnetic wave is given by (The speed of light $= 3\times10^{8}\mathrm{m/s}$)

$$\mathbf{B = 100\times10^{-6}\sin\left[2\pi\times2\ \times10^{15}\left(t-\frac{x}{c}\right)\right]}$$

then the maximum electric field associated with it is:
[10 Jan. 2019 I]

Options:

A. $6\times10^{4}\mathrm{N/C}$

B. $3\times10^{4}\mathrm{N/C}$

C. $4\times10^{4}\mathrm{N/C}$

D. $4.510^{4}\sim\mathrm{N/C}$

Answer: B

Solution:

Using, formula $E_0 = B_0\times C$
$= 100\times10^{-6}\times3\times10^{8}$
$= 3\times10^{4}\mathrm{N/C}$
Here we assumed that
$B_0 = 100\times10^{-6}$ is in tesla (T) units

Question98

The electric field of a plane polarized electromagnetic wave in free space at time t = 0 is given by an expression

$\vec{E}(x, y) = 10\hat{j}\cos[(6x + 8z)]$

The magnetic field $\vec{B}(x, z, t)$ is given by: (c is the velocity of light) [10 Jan 2019, II]

Options:

A. $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x - 8z + 10ct)]$

B. $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

C. $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

D. $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x + 8z + 10ct)]$

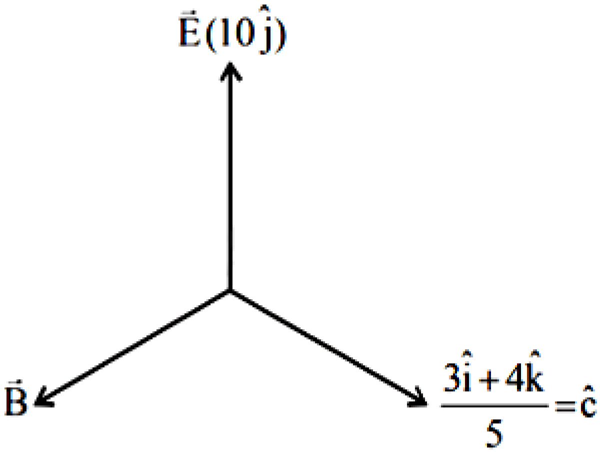
Answer: B

Solution:

$E = 10\hat{j}\cos\left[\left(6\hat{i} + 8\hat{k}\right) . \left(x\hat{i} + z\hat{k}\right)\right]$

$= 10\hat{j}\cos\left[\vec{K} . \vec{r}\right]$

$\therefore \vec{K} = 6\hat{i} + 8\hat{K}$; direction of waves travel
i. e. direction of 'c'



$\hat{C} \times \hat{E} = \frac{-4\hat{i} + 3\hat{k}}{5}$

$\vec{B} = \frac{E}{C} = \frac{10}{C}$

$\therefore \vec{B}\,\frac{10}{C}\left(\frac{-4\hat{i} + 3\hat{k}}{5}\right) = \left(\frac{-8\hat{i} + 6\hat{k}}{C}\right)$

or, magnetic field $\vec{B}(x, z, t) = \frac{1}{C}$

$\left(6\hat{k} - 8\hat{i}\right) \cos(6x + 8z - 10ct)$

Question99

An EM wave from air enters a medium. The electric fields are $\vec{E}_1 = E_{01}\hat{x}\cos\left[2\pi v\left(\frac{z}{c} - t\right)\right]$ in air and $\vec{E}_2 = E_{02}\hat{x}\cos[k(2z - ct)]$ in medium, where the wave number k and frequency v refer to their values in air. The medium is nonmagnetic. If $\in_{r_1}$ and $\in_{r_2}$ refer to relative permittivities of air and medium respectively, which of the following options is correct? [9 Jan 2019, I]

Options:

A. $\frac{\in_{r_1}}{\in_{r_2}} = 4$

B. $\frac{\in_{r_1}}{\in_{r_2}} = 2$

C. $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{4}$

D. $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{2}$

Answer: C

Solution:

Velocity of EM wave is given by $v = \frac{1}{\sqrt{\mu \in}}$

Velocity in air $= \frac{\omega}{k} = C$

Velocity in medium $= \frac{C}{2}$

Here, $\mu_1 = \mu_2 = 1$ as medium is non-magnetic

$$\therefore \frac{\frac{1}{\sqrt{\epsilon_{r_1}}}}{\frac{1}{\sqrt{\epsilon_{r_2}}}} = \frac{C}{\left(\frac{C}{2}\right)} = 2 \Rightarrow \frac{\epsilon_{r_1}}{\epsilon_{r_2}} = \frac{1}{4}$$

Question100

The energy associated with electric field is (U_E) and with magnetic fields is (U_B) for an electromagnetic wave in free space. Then : [9 Jan 2019, II]

Options:

A. $U_E = \frac{U_B}{2}$

B. $U_E > U_B$

C. $U_E < U_B$

D. $U_E = U_B$

Answer: D

Solution:

Average energy density of magnetic field,

$$u_B = \frac{B_0^{\,2}}{4\mu_0}$$

Average energy density of electric field,

$$u_E = \frac{\varepsilon_0 E_0^{\,2}}{4}$$

Now, $E_0 = CB_0$ and $C^2 = \frac{1}{\mu_0 \epsilon_0}$

$$u_E = \frac{\epsilon_0}{4} \times C^2 B_0^{\,2} = \frac{\varepsilon_0}{4} \times \frac{1}{\mu_0 \varepsilon_0} \times B_0^{\,2} = \frac{B_0^{\,2}}{4\mu_0} = u_B$$

$\therefore u_E = u_B$

Since energy density of electric and magnetic field is same, so energy associated with equal volume will be equal i.e. $u_E = u_B$

Question101

Given below in the left column are different modes of communication using the kinds of waves given in the right column.

A. Optical Fibre Communication	P. Ultrasound
B. Radar	Q. Infrared Light
C. Sonar	R. Microwaves
D. Mobile Phones	S. Radio Waves

From the options given below, find the most appropriate match between entries in the left and the right column.
[10 April 2019, I]

Options:

A. A – Q, B – S, C – R, D – P

B. A – S, B – Q, C – R, D – P

C. A – Q, B – S, C – P, D – R

D. A – R, B – P, C – S, D – Q

Answer: C

Solution:

Optical Fibre Communication - Infrared Light
Radar-Radio Waves
Sonar-Ultrasound
Mobile Phones - Microwaves

Question102

An electromagnetic wave is represented by the electric field $\vec{E} = E_0 \hat{n} \sin[\omega t + (6y - 8z)]$. Taking unit vectors inx, y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propogation $\hat{s}$ is :
[12 April 2019, I]

Options:

A. $\hat{s} = \frac{3\hat{i} - 4\hat{j}}{5}$

B. $\hat{s} = \frac{-4\hat{k} + 3\hat{j}}{5}$

C. $\hat{s} = \left(\frac{-3\hat{j} + 4\hat{k}}{5}\right)$

D. $\hat{s} = \frac{3\hat{j} - 3\hat{k}}{5}$

Answer: C

Solution:

$$\hat{S} = \frac{6\hat{j} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{-3\hat{j} + 4\hat{k}}{5}$$

Question103

A plane electromagnetic wave having a frequency v = 23.9 GHz propagates along the positive z -direction in free space. The peak value of the Electric Field is 60V ⁄ m. Which among the following is the acceptable magnetic field component in the electromagnetic wave? [12 April 2019, II]

Options:

A. $\vec{B} = 2 \times 10^7 \sin(0.5 \times 10^3 z + 1.5 \times 10^{11} t)\hat{i}$

B. $\vec{B} = 2 \times 10^7 \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t)\hat{i}$

C. $\vec{B} = 60 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)\hat{k}$

D. $\vec{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x + 0.5 \times 10^{11} t)\hat{j}$

Answer: B

Solution:

$$B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8}$$
$$= 20 \times 10^{-8} T = 2 \times 10^{-7} T$$
$$K = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi \times 23.9 \times 10^9}{3 \times 10^8} = 500$$

Therefore, $\vec{B} = B_0 \sin(kz - \omega t)$

$$= 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t)\, i$$

Question104

The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \hat{i} \cos(kz) \cos(\omega t)$

The corresponding magnetic field is then given by : [10 April 2019, I]

Options:

A. $\vec{B} = \frac{E_0}{C} \hat{j} \cos(kz) \cos(\omega t)$

B. $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \cos(\omega t)$

C. $\vec{B} = \frac{E_0}{C} \hat{j} \cos(kz) \sin(\omega t)$

D. $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \cos(\omega t)$

Answer: A

Solution:

$$\frac{E_0}{B_0} = C$$
$$\Rightarrow B_0 = \frac{E_0}{C}$$

Given that $\vec{E} = E_0 \cos(kz)\cos(\omega t)\,\vec{i}$

$$\vec{E} = \frac{E_0}{2}\left[\cos(kz-\omega t)\,\hat{i} - \cos(kz+\omega t)\,\hat{i}\right]$$

Correspondingly

$$\vec{B} = \frac{B_0}{2}\left[\cos(kz-\omega t)\,\hat{j} - \cos(kz+\omega t)\,\hat{j}\right]$$

$$\vec{B} = \frac{B_0}{2} \times 2\sin kz \sin\omega t$$

$$\vec{B} = \left(\frac{E_0}{C}\sin kz \sin\omega t\right)\hat{j}$$

Question105

Light is incident normally on a completely absorbing surface with an energy flux of 25W cm^{-2}. If the surface has an area of 25cm^{2}, the momentum transferred to the surface in 40 min time duration will be:
[10 April 2019, II]

Options:

A. 6.3×10^{-4}N s

B. 1.4×10^{-6}N s

C. 5.0×10^{-3}N s

D. 3.5×10^{-6}N s

Answer: C

Solution:

Pressure, $P = \frac{I}{C}$

$$\Rightarrow \frac{F}{A} = \frac{I}{C}$$

$$\Rightarrow F = \frac{IA}{C} = \frac{\Delta p}{\Delta t}$$

$$\Rightarrow \Delta p = \frac{I}{C} A \Delta t$$

$$= \frac{(25 \times 25) \times 10^{4} \times 10^{-4} \times 40 \times 60}{3 \times 10^{8}} N - s$$

$= 5 \times 10^{-3} N - s$

Question106

The magnetic field of a plane electromagnetic wave is given by:
$\vec{B} = B_0 \hat{i}[\cos(kz - \omega t)] + B_1 \hat{j} \cos(kz + \omega t)$
Where $B_0 = 3 \times 10^{-5} T$ and $B_1 = 2 \times 10^{-6} T$
The rms value of the force experienced by a stationary charge $Q = 10^{-4} C$ at $z = 0$ is closest to:
[9 April 2019 I]

Options:

A. 0.6 N

B. 0.1 N

C. 0.9 N

D. 3×10^{-2} N

Answer: A

Solution:

$B_0 = \sqrt{B_0{}^2 + B_1{}^2} = \sqrt{30^2 + 2^2} \times 10^{-6} = 30 \times 10^{-6} T$
$\therefore E_0 = CB = 3 \times 10^8 \times 30 \times 10^{-6}$
$= 9 \times 10^3 V/m$
$\frac{E_0}{V_2} = \frac{9}{\sqrt{2}} \times 10^3 V/m$
Force on the charge,
$F = EQ = \frac{9}{\sqrt{2}} \times 10^3 \times 10^{-4} \simeq 0.64 N$

Question107

A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x - direction. At a particular point in space

and time, $\vec{E} = 6.3\hat{j}V/m$. The corresponding magnetic field $\vec{B}$, at that point will be:
[9 April 2019 I]

Options:

A. $18.9 \times 10^{-8}\hat{k}T$

B. $2.1 \times 10^{-8}\hat{k}T$

C. $6.3 \times 10^{-8}\hat{k}T$

D. $18.9 \times 10^{8}\hat{k}T$

Answer: B

Solution:

As we know,

$$|\vec{B}| = \frac{|\vec{E}|}{C} = \frac{6.3}{3 \times 10^8} = 2.1 \times 10^{-8}T$$

and $\hat{E} \times \hat{B} = \hat{C}$

$\hat{J} \times \hat{B} = \hat{i}$ [$\because$ E M wave travels along +(ve)x -direction.]

$\therefore \hat{B} = \hat{k}$ or $\vec{B} = 2.1 \times 10^{-8}\hat{k}T$

Question108

$50W/m^2$ energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on $1m^2$ surface area will be
[9 April 2019, II]

Options:

A. $15 \times 10^{-8}N$

B. $20 \times 10^{-8}N$

C. $10 \times 10^{-8} \mathrm{N}$

D. $35 \times 10^{-8} \mathrm{N}$

Answer: B

Solution:

$$
\begin{aligned}
F &= (1+r)\frac{IA}{C} \\
&= \frac{(1+0.25) \times 50 \times 1}{3 \times 10^{8}} \\
&\simeq 20 \times 10^{-8} \mathrm{N}
\end{aligned}
$$

Question109

A plane electromagnetic wave travels in free space along the x -direction. The electric field component of the wave at a particular point of space and time is $E = 6 \mathrm{V\,m}^{-1}$ along y -direction. Its corresponding magnetic field component, B would be:
[8 April 2019 I]

Options:

A. $2 \times 10^{-8} \mathrm{T}$ along z -direction

B. $6 \times 10^{-8} \mathrm{T}$ along x -direction

C. $6 \times 10^{-8} \mathrm{T}$ along z -direction

D. $2 \times 10^{-8} \mathrm{T}$ along y -direction

Answer: A

Solution:

The relation between amplitudes of electric and magnetic field in free space is given by

$$B_0 = \frac{E_0}{c} = \frac{6}{3 \times 10^{8}} = 2 \times 10^{-8} \mathrm{T}$$

Propagation direction $= \hat{E} \times \hat{B}$

$\hat{i} = \hat{j} \times \hat{B}$
$\Rightarrow \hat{B} = \hat{k}$
∴ The magnetic field component will be along z direction.

Question110

The magnetic field of an electromagnetic wave is given by:

$$\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t) \left(2\hat{i} + \hat{j}\right) \frac{Wb}{m^2}$$

The associated electric field will be :
[8 April 2019 II]

Options:

A. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t)\left(2\hat{i} + \hat{j}\right) \frac{V}{m}$

B. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t)\left(-2\hat{j} + \hat{i}\right) \frac{V}{m}$

C. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t)\left(-\hat{i} + 2\hat{j}\right) \frac{V}{m}$

D. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t)\left(\hat{i} - 2\hat{j}\right) \frac{V}{m}$

Answer: C

Solution:

$E_0 = cB_0 = 3 \times 10^8 \times 1.6 \times 10^{-6} = 4.8 \times 10^2 V/m$
Also $\vec{S} \Rightarrow \vec{E} \times \vec{B}$
or $-\vec{K} \Rightarrow \vec{E} \times \left(2\hat{i} + \hat{j}\right)$
Therefore direction of $\vec{E} \rightarrow \left(-\hat{i} + 2\hat{j}\right)$

Question111

A plane electromagnetic wave of wavelength l has an intensity I. It is propagating along the positive Y– direction. The allowed expressions

for the electric and magnetic fields are given by
[Online April 16, 2018]

Options:

A. $\vec{E} = \sqrt{\frac{I}{\varepsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{i}; \ \vec{B} = \frac{1}{c}E\hat{k}$

B. $\vec{E} = \sqrt{\frac{I}{\varepsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \ \vec{B} = -\frac{1}{c}E\hat{i}$

C. $\vec{E} = \sqrt{\frac{2I}{\varepsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \ \vec{B} = +\frac{1}{c}E\hat{i}$

D. $\vec{E} = \sqrt{\frac{2I}{\varepsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y + ct)\right] \hat{k}; \vec{B} = \frac{1}{c}E\hat{i}$

Answer: C

Solution:

If E_0 is magnitude of electric field then

$\frac{1}{2}\varepsilon_0 E^2 \times C = 1 \Rightarrow E_0 = \sqrt{\frac{2I}{C\varepsilon_0}}$

$E_0 = \frac{E_0}{C}$

Direction of $\vec{E} \times \vec{B}$ will be along $+\hat{j}$

Question112

A monochromatic beam of light has a frequency$v = \frac{3}{2\pi} \times 10^{12}Hz$ and is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$. It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is:
[Online April 15, 2018]

Options:

A.

$$k\frac{E_0}{C}\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right)\cos\left[10^4\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right).\vec{r}-(3\times10^{12})t\right]$$

B.

$$\frac{E_0}{C}\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right)\cos\left[10^4\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right).\vec{r}-(3\times10^{12})t\right]$$

C.

$$\frac{E_0}{C}\hat{k}\cos\left[10^4\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right).\vec{r}-(3\times10^{12})t\right]$$

D.

$$\frac{E_0}{C}\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{\sqrt{3}}\cos\left[10^4\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right).\vec{r}+(3\times10^{12})t\right]$$

Answer: C

Solution:

$\hat{E}\times\hat{B}$ should give the direction of wave propagation

$$\Rightarrow\hat{K}\times\hat{B}\parallel\frac{\hat{i}\times\hat{j}}{\sqrt{2}}\Rightarrow\hat{K}\times\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)=\frac{\hat{j}-\left(-\hat{i}\right)}{\sqrt{2}}=\frac{\hat{i}+\hat{j}}{\sqrt{2}}\parallel\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

Option (a), option (b) and option (d) does not satisfy.

Wave propagation vector $\hat{K}$ should along $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Question113

The electric field component of a monochromatic radiation is given by $\vec{E}=2E_0\hat{i}\cos kz\cos\omega t$Its magnetic field $\vec{B}$ is then given by : [Online April 9, 2017]

Options:

A. $\frac{2E_0}{c}\hat{j}\sin kz\cos\omega t$

B. $-\frac{2E_0}{c}\hat{j}\sin kz\cos\omega t$

C. $\frac{2E_0}{c}\hat{j}\sin kz\cos\omega t$

D. $\frac{2E_0}{c}\hat{j}\cos kz\cos\omega t$

Answer: C

Solution:

Given, Electric field component of monochromatic radiation, $(\vec{E}) = 2E_0\hat{i}\cos kz\cos\omega t$

We know that, $\frac{dE}{dz} = -\frac{dB}{dt}$

$\frac{dE}{dz} = -2E_0k\sin kz\cos\omega t = -\frac{dB}{dt}$

$dB = +2E_0k\sin kz\cos\omega t\,dt$ .....(i)

Integrating eq^n (i), we have

$B = +2E_0k\sin kz\int\cos\omega t\,dt$

Magnetic field is given by,

$= +2E_0\frac{k}{\omega}\sin kz\sin\omega t$

We also know that,

$\frac{E_0}{B_0} = \frac{\omega}{k} = c$

Magnetic field vector,

$\vec{B} = \frac{2E_0}{c}\hat{j}\sin kz\cos\omega t$

Question114

Magnetic field in a plane electromagnetic wave is given by $\vec{B} = B_0\sin(kx+\omega t)\hat{j}$ T Expression for corresponding electric field will be : Where c is speed of light. [Online April 8,2017]

Options:

A. $\vec{E} = B_0c\sin(kx+\omega t)\hat{k}$ V / m

B. $\vec{E} = \frac{B_0}{c}\sin(kx+\omega t)\hat{k}$ V / m

C. $\vec{E} = -B_0 c \sin(kx + \omega t) \hat{k}\ V/m$

D. $\vec{E} = B_0 c \sin(kx - \omega t) \hat{k}\ V/m$

Answer: A

Solution:

Speed of EM wave in force space $(c) = \frac{E_0}{B_0}$

or $\vec{E} = cB_0 \sin(kx + \omega t) \hat{k}$

Question115

Consider an electromagnetic wave propagating in vacuum. Choose the correct statement:
[Online April 10, 2016]

Options:

A. For an electromagnetic wave propagating in +y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{z}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_z(x, t) \vec{y}$

B. For an electromagnetic wave propagating in +y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{y}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x, t) \hat{z}$

C. For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(y, z, t)\left(\hat{y} + \hat{z}\right)$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(y, z, t)\left(\hat{y} + \hat{z}\right)$

D. For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t)\left(\hat{y} - \hat{z}\right)$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x, t)\left(\hat{y} + \hat{z}\right)$

Answer: D

Solution:

Wave in X-direction means E and B should be function of x and t
$\hat{y}-\hat{z} \perp \hat{y}+\hat{z}$

Question116

Arrange the following electromagnetic radiations per quantum in the order of increasing energy :
A : Blue light
B : Yellow light
C : X-ray
D : Radio wave.
[2016]

Options:

A. C, A, B, D

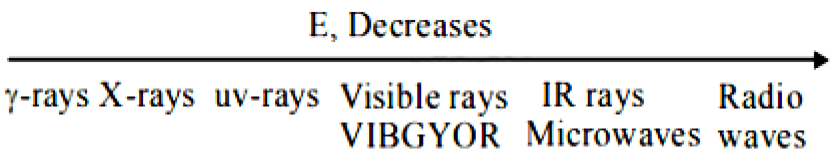
B. B, A, D, C

C. D, B, A, C

D. A, B, D, C

Answer: C

Solution:



Radio wave < yellow light < blue light $ (Increasing order of energy)

Question117

Microwave oven acts on the principle of :
[Online April 9, 2016]

Options:

A. giving rotational energy to water molecules

B. giving translational energy to water molecules

C. giving vibrational energy to water molecules

D. transferring electrons from lower to higher energy levels in water molecule

Answer: C

Solution:

Microwave oven acts on the principle of giving vibrational energy to water molecules.

Question118

For plane electromagnetic waves propagating in the z-direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively?
[Online April 11, 2015]

Options:

A. $\left(2\hat{i}+3\hat{j}\right)$ and $\left(\hat{i}+2\hat{j}\right)$

B. $\left(-2\hat{i}-3\hat{j}\right)$ and $\left(3\hat{i}-2\hat{j}\right)$

C. $\left(\hat{i}+2\hat{j}\right)$ and $\left(2\hat{i}-\hat{j}\right)$

D. $\left(3\hat{i}+4\hat{j}\right)$ and $\left(4\hat{i}-3\hat{j}\right)$

Answer: B

Solution:

As we know, $\vec{E}\,.\,\vec{B}=0 \because \left[\vec{E}\perp\vec{B}\right]$
and $\vec{E}\times\vec{B}$ should be along $\hat{Z}$ direction
As $\left(-2\hat{i}-3\hat{j}\right)\times\left(3\hat{i}-2\hat{j}\right)=5\hat{k}$
Hence option (b) is the correct answer.

Question119

An electromagnetic wave travelling in the x -direction has frequency of 2×10^{14} Hz and electric field amplitude of $27\ V\ m^{-1}$. From the options given below, which one describes the magnetic field for this wave ? [Online April 10,2015]

Options:

A. $\vec{B}(x, t) = (3 \times 10^{-8} T)\hat{j}\ \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)$

B. $\vec{B}(x, t) = (9 \times 10^{-8} T)\hat{i}\ \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)$

C. $\vec{B}(x, t) = (9 \times 10^{-8} T)\hat{j}\ \sin[(1.5 \times 10^{-8} x - 2 \times 10^{14} t)$

D. $\vec{B}(x, t) = (9 \times 10^{-8} T)\hat{k}\ \sin[2\pi(1.5 \times 10^{-6} x - 2 \times 10^{14} t)$

Answer: D

Solution:

As we know,

$B_0 = \frac{E_0}{C} = \frac{27}{3 \times 10^8} = 9 \times 10^{-8}$ tesla

Oscillation of B can be only along $\hat{j}$ or $\hat{k}$ direction. $\omega = 2\pi f = 2\pi \times 2 \times 10^{14} Hz$

$\therefore \vec{B}(x, t) = (9 \times 10^{-8} T)\hat{k}\ \sin[2\pi(1.5 \times 10^{-6} \times - 2 \times 10^{4} t)]$

Question120

During the propagation of electromagnetic waves in a medium: [2014]

Options:

A. Electric energy density is double of the magnetic energy density.

B. Electric energy density is half of the magnetic energy density.

C. Electric energy density is equal to the magnetic energy density.

D. Both electric and magnetic energy densities are zero

Answer: C

Solution:

$E_0 = CB_0$ and $C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

Electric energy density $= \frac{1}{2}\varepsilon_0 E_0^{\ 2} = \mu_E$

Magnetic energy density $= \frac{1}{2}\frac{Bo^2}{\mu_0} = \mu_B$

Thus, $\mu_E = \mu_B$

Energy is equally divided between electric and magnetic field

----

Question121

A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 5 m from the lamp will be nearly: [Online April 12, 2014]

Options:

A. 1.34 V/m

B. 2.68 V/m

C. 4.02 V/m

D. 5.36 V/m

Answer: B

Solution:

Wavelength of monochromatic green light $= 5.5 \times 10^{-5}\text{cm}$

Intensity $I = \frac{\text{Power}}{\text{Area}}$

$= \frac{100 \times (3/100)}{4\pi(5)^2} = \frac{3}{100\pi}\text{W m}^{-2}$

Now, half of this intensity (I) belongs to electric field and half of that to magnetic field, therefore,

$\frac{I}{2} = \frac{1}{4}\varepsilon_0 E_0^{\,2} C$

or $E_0 = \sqrt{\frac{2I}{\varepsilon_0 C}}$

$$= \sqrt{\frac{2 \times \left(\frac{3}{100}\pi\right)}{\left(\frac{1}{4\pi \times 9 \times 10^9}\right) \times (3 \times 10^8)}}$$

$= \sqrt{\frac{6}{25}} \times 30 = \sqrt{7.2}$

$\therefore E_0 = 2.68\text{V}/\text{m}$

Question122

An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is $4\text{V}/\text{m}$. If $\varepsilon_0 = 8.8 \times 10^{-12}\text{C}^2/\text{N}-\text{m}^2$, then average energy density of electric field will be:

[Online April 11,2014]

Options:

A. $35.2 \times 10^{-10}\text{J}/\text{m}^3$

B. $35.2 \times 10^{-11}\text{J}/\text{m}^3$

C. $35.2 \times 10^{-12}\text{J}/\text{m}^3$

D. $35.2 \times 10^{-13}\text{J}/\text{m}^3$

Answer: C

Solution:

Given: Amplitude of electric field,

$E_0 = 4\text{v}/\text{m}$

Absolute permitivity, $\varepsilon_0 = 8.8 \times 10^{-12} c^2 / N - m^2$
Average energy density $u_E = ?$
Applying formula,
Average energy density $u_E = \frac{1}{4}\varepsilon_0 E^2$
$\Rightarrow u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2$
$= 35.2 \times 10^{-12} J / m^3$

Question123

Match List - I (Electromagnetic wave type) with List - II (Its association/application) and select the correct option from the choices given below the lists:

List 1	List 2
1. Infrared waves	(i) To treat muscular strain
2. Radio waves	(ii) For broadcasting
3. X-rays	(iii) To detect fracture of bones
4. Ultraviolet rays	(iv) Absorbed by the ozone layer of the atmosphere

[2014]

Options:

A. 1-(iv), 2-(iii), 3-(ii), 4-(i)

B. 1-(i), 2-(ii), 3-(iv), 4-(iii)

C. 1-(iii), 2-(ii), 3-(i), 4-(iv)

D. 1-(i), 2-(ii), 3-(iii), 4-(iv)

Answer: D

Solution:

(1) Infrared rays are used to treat muscular strain because these are heat rays.
(2) Radio waves are used for broadcasting because these waves have very long wavelength ranging from few centimeters to few hundred kilometers.

(3) X-rays are used to detect fracture of bones because they have high penetrating power but they can't penetrate through denser medium like dones.
(4) Ultraviolet rays are absorbed by ozone of the atmosphere.

Question124

If microwaves, X rays, infrared, gamma rays, ultra-violet,radio waves and visible parts of the electromagnetic spectrum are denoted by M, X, I, G, U, R and V then which of the following is the arrangement in ascending order of wavelength ? [Online April 19, 2014]

Options:

A. R, M, I, V, U, X and G

B. M, R, V, X, U, G and I

C. G, X, U, V, I, M and R

D. I, M, R, U, V, X and G

Answer: C

Solution:

(c) Gamma rays $<\mathrm{X}$ -rays < Ultra violet < Visible rays < Infrared rays < Microwaves < Radio waves.

Question125

Match the List-I (Phenomenon associated with electromagnetic radiation) with List-II (Part of electromagnetic spectrum) and select the correct code from the choices given below this lists:

	List I		List II
I	Doublet of sodium	(A)	Visible radiation
II	Wavelength corresponding to temperature associated with the isotropic radiation filling all space	(B)	Microwave
III	Wavelength emitted by atomic hydrogen in interstellar space	(C)	Short radio wave
IV	Wavelength of radiation arising from two close energy levels in hydrogen	(D)	X-rays

[Online April 11, 2014]

Options:

A. (I)-(A), (II)-(B), (III)-(B), (IV)-(C)

B. (I)-(A), (II)-(B), (III)-(C), (IV)-(C)

C. (I)-(D), (II)-(C), (III)-(A), (IV)-(B)

D. (I)-(B), (II)-(A), (III)-(D), (IV)-(A)

Answer: D

Solution:

Wavelength emitted by atomic hydrogen in interstellar space - Part of short radio wave of electromagnetic spectrum.
Doublet of sodium - visible radiation.

Question126

Match List I (Wavelength range of electromagnetic spectrum) with List II (Method of production of these waves) and select the correct option from the options given below the lists.

	List I		List II
(1)	700 nm to 1 mm	(i)	Vibration of atoms and molecules.
(2)	1 nm to 400 nm	(ii)	Inner shell electrons in atoms moving from one energy level to a lower level.
(3)	$<10^{-3}$ nm	(iii)	Radioactive decay of the nucleus.
(4)	1 mm to0.1 m	(iv)	Magnetron valve.

[Online April 9, 2014]

Options:

A. (1)-(iv), (2)-(iii), (3)-(ii), (4)-(i)

B. (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii)

C. (1)-(ii), (2)-(iii), (3)-(iv), (4)-(i)

D. (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)

Answer: D

Solution:

Vibration of atoms and molecules $700\mathrm{nm}$ to $1\mathrm{mm}$
Radioactive decay of the nucleus $<10^{-3}\mathrm{nm}$
Magnetron valve $1\mathrm{mm}$ to $0.1\mathrm{m}$

Question127

The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength is : [2013]

Options:

A. 3 V/m

B. 6 V/m

C. 9 V/m

D. 12 V/m

Answer: B

Solution:

From question,
$B_0 = 20\mathrm{nT} = 20 \times 10^{-9}\mathrm{T}$
($\because$ velocity of light in vacuum $C = 3 \times 10^{8}\mathrm{ms}^{-1}$)
$\vec{E}_0 = \vec{B}_0 \times \vec{C}$

$\left|\vec{E}_0\right| = \left|\vec{B}\right| . \left|\vec{C}\right| = 20 \times 10^{-9} \times 3 \times 10^{8}$
$= 6V / m$

Question128

A plane electromagnetic wave in a non-magnetic dielectric medium is given by $\vec{E} = \vec{E}_0(4 \times 10^{-7}x - 50t)$ with distance being in meter and time in seconds. The dielectric constant of the medium is : [Online April 22, 2013]

Options:

A. 2.4

B. 5.8

C. 8.2

D. 4.8

Answer: B

Question129

Select the correct statement from the following : [Online April 9, 2013]

Options:

A. Electromagnetic waves cannot travel in vacuum.

B. Electromagnetic waves are longitudinal waves.

C. Electromagnetic waves are produced by charges moving with uniform velocity.

D. Electromagnetic waves carry both energy and momentum as they propagate through space.

Answer: D

Solution:

Electromagnetic waves do not required any medium to propagate. They can travel in vacuum. They are transverse in nature like light. They carry both energy and momentum.
A changing electric field produces a changing magnetic field and vice-versa. Which gives rise to a transverse wave known as electromagnetic wave.

Question130

Photons of an electromagnetic radiation has an energy 11 keV each. To which region of electromagnetic spectrum does it belong ? [Online April 9, 2013]

Options:

A. X-ray region

B. Ultra violet region

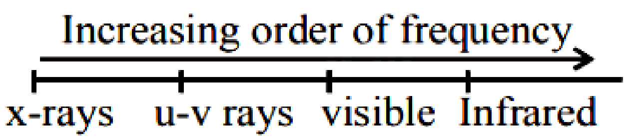
C. Infrared region

D. Visible region

Answer: A

Solution:

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$
$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{11 \times 1000 \times 1.6 \times 10^{-19}}$$
$$= 12.4\ \text{Å}$$



wavelength range of visible region is 4000Å to 7800Å.

Question131

An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by vec X and that of wave propagation by $\vec{k}$. Then
[2012]

Options:

A. $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B} \times \vec{E}$

B. $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E} \times \vec{B}$

C. $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E} \times \vec{B}$

D. $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B} \times \vec{E}$

Answer: B

Solution:

∵ The E.M. wave are transverse in nature i.e.,

$= \frac{\vec{k} \times \vec{E}}{\mu} = \vec{H}$ .....(i)

where $\vec{H} = \frac{\vec{B}}{\mu}$

and $\frac{\vec{k} \times \vec{H}}{\omega\varepsilon} = -\vec{E}$ ....(ii)

$\vec{k}$ is $\perp \vec{H}$ and $\vec{k}$ is also $\perp$ to $\vec{E}$

The direction of wave propagation is parallel to $\vec{E} \times \vec{B}$.
The direction of polarization is parallel to electric field.

Question132

An electromagnetic wave with frequency ω and wavelength λ travels in the + y direction. Its magnetic field is along + x-axis. The vector equation for the associated electric field (of amplitude E_0) is

[Online May 19, 2012]

Options:

A. $\overrightarrow{E} = -E_0 \cos\left(\omega t + \frac{2\pi}{\lambda} y\right) \hat{x}$

B. $\overrightarrow{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{x}$

C. $\overrightarrow{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$

D. $\overrightarrow{E} = -E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$

Answer: C

Solution:

In an electromagnetic wave electric field and magnetic field are perpendicular to the direction of propagation of wave. The vector equation for the electric field is

$\overrightarrow{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$

Question133

The frequency of X-rays; γ-rays and ultraviolet rays are respectively a, b and c then
[Online May 26, 2012]

Options:

A. $a < b; b > c$

B. $a > b ; b > c$

C. $a < b < c$

D. $a = b = c$

Answer: A

Solution:

Frequency range of γ -ray,
$b = 10^{18} - 10^{23} Hz$
Frequency range of X-ray,
$a = 10^{16} - 10^{20} Hz$
Frequency range of ultraviolet ray,
$c = 10^{15} - 10^{17} Hz$
$\therefore a < b; b > c$

Question134

An electromagnetic wave of frequency v = 3.0 MHz passes from vacuum into a dielectric medium with permittivity ∈ = 4.0 . Then [2004]

Options:

A. wave length is halved and frequency remains unchanged

B. wave length is doubled and frequency becomes half

C. wave length is doubled and the frequency remains unchanged

D. wave length and frequency both remain unchanged.

Answer: A

Solution:

Frequency remains unchanged during refraction Velocity of EM wave in vacuum

$$V_{vacuum} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = C$$

$$v_{med} = \frac{1}{\sqrt{\mu_0 \epsilon_0 \times 4}} = \frac{c}{2}$$

$$\frac{\lambda_{med}}{\lambda_{vacuum}} = \frac{v_{med}}{v_{vacuum}} = \frac{c/2}{c} = \frac{1}{2}$$

∴ Wavelength is halved and frequency remains unchanged

Question135

Electromagnetic waves are transverse in nature is evident by [2002]

Options:

A. polarization

B. interference

C. reflection

D. diffraction

Answer: A

Solution:

The phenomenon of polarisation is shown only by transverse waves. The vibration of electromagnetic wave are restricted through polarization in a direction perpendicular to wave propagation.
